

Supplementary materials for the paper “Integrated Volatility Matrix Estimation with Nonparametric Eigenvalue Regularization” - Proof of Theorem 1,2,3 and 4.

We introduce some notations first. If we assume the jump-diffusion model (4.2) for the log-price process $\mathbf{X}_t$, and all estimators are constructed using the jumps-removed data

$$\tilde{\mathbf{Y}}_t = \mathbf{Y}_t - \hat{\mathbf{J}}_t = (\mathbf{X}_t - \hat{\mathbf{J}}_t) + \boldsymbol{\epsilon}_t = \tilde{\mathbf{X}}_t + \boldsymbol{\epsilon}_t, \quad (\text{S.1})$$

where $\{\hat{\mathbf{J}}_t\}$ is the estimated jump process using the wavelet method in Fan and Wang (2007), then $\{\tilde{\mathbf{X}}_t\}$ represents the jumps-removed log-price process. For $j = 1, \dots, L$ and $v_s = v_s^j$ for $s = 0, \dots, n(j)$, we then have

$$\tilde{\mathbf{Y}}(s) = \tilde{\mathbf{X}}(s) + \boldsymbol{\epsilon}(s) = \tilde{\mathbf{X}}_{v_s} + \mathbf{E}(s),$$

where we define

$$\mathbf{E}(s) = \boldsymbol{\epsilon}(s) + \tilde{\mathbf{X}}(s) - \tilde{\mathbf{X}}_{v_s} = \boldsymbol{\epsilon}(s) + (\mathbf{X}(s) - \hat{\mathbf{J}}(s)) - (\mathbf{X}_{v_s} - \hat{\mathbf{J}}_{v_s}).$$

Replacing $\mathbf{X}(s)$ and $\mathbf{X}_{v_s}$ by $\mathbf{X}(s) + \mathbf{J}(s)$ and $\mathbf{X}_{v_s} + \mathbf{J}_{v_s}$ respectively where $\mathbf{X}_t$ is the vector of pure jump log-price process, we then have

$$\mathbf{E}(s) = \boldsymbol{\epsilon}(s) + \tilde{\mathbf{X}}(s) - \tilde{\mathbf{X}}_{v_s} = \boldsymbol{\epsilon}(s) + (\mathbf{X}(s) - \mathbf{X}_{v_s}) + (e(\mathbf{J}(s)) - e(\mathbf{J}_{v_s})),$$

where $e(\mathbf{J}_t) = \mathbf{J}_t - \hat{\mathbf{J}}_t$. Hereafter, we denote $\mathbf{E}(s)$ by the above expansion, and $\tilde{\mathbf{X}}_t = \mathbf{X}_t + e(\mathbf{J}_t)$.

Before we present the proofs of our theorems, we introduce five important lemmas first. The first two lemmas come from Lam and Feng (2018) and are used for proving Theorem 1, 2 and 3. The remaining three lemmas are developed for proving Theorem 2 and 3.

For any integer $m \geq 1$, define

$$\begin{aligned} [\tilde{\mathbf{X}}_v, \tilde{\mathbf{X}}_v^T]_j^{(m)} &= \frac{1}{m} \sum_{s, s+m \in S^j(m)} (\tilde{\mathbf{X}}_{v_{s+m}} - \tilde{\mathbf{X}}_{v_s})(\tilde{\mathbf{X}}_{v_{s+m}} - \tilde{\mathbf{X}}_{v_s})^T, \\ [\tilde{\mathbf{X}}_v, \mathbf{E}^T]_j^{(m)} &= \frac{1}{m} \sum_{s, s+m \in S^j(m)} (\tilde{\mathbf{X}}_{v_{s+m}} - \tilde{\mathbf{X}}_{v_s})(\mathbf{E}(s+m) - \mathbf{E}(s))^T, \\ [\mathbf{E}, \mathbf{E}^T]_j^{(m)} &= \frac{1}{m} \sum_{s, s+m \in S^j(m)} (\mathbf{E}(s+m) - \mathbf{E}(s))(\mathbf{E}(s+m) - \mathbf{E}(s))^T. \end{aligned} \quad (\text{S.2})$$

Lemma 1 *Let all the assumptions in Theorem 4 hold. Then with $p/n \rightarrow c \geq 0$, we have*

$$\max_{\substack{i=1, \dots, p \\ j=1, \dots, L}} \left| \frac{\mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \tilde{\mathbf{X}}_v^T]_j^{(K_m)} \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1 \right| = O_P(K_m^{1/2} n^{-1/2} + p_f^{1/2} n^{-1/2}).$$

The proof for Lemma 1 can be found in the intermediate results in the proof of Lemma 1 in Lam and Feng (2018). We do not repeat their calculations here.

Lemma 2 *Let all the assumptions in Theorem 4 hold. Then with $p/n \rightarrow c \geq 0$, we have*

$$\max_{\substack{i=1, \dots, p \\ j=1, \dots, L}} \max_{s=1, \dots, n(j)} \left| \frac{\mathbf{p}_{ij}^T (\mathbf{X}_{v_s} - \mathbf{X}(s))}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} \right| = O_P(n^{-1/2}).$$

This Lemma is in fact the same as Lemma 2 in Lam and Feng (2018) where $\alpha = 1/6$ there. The different result stems from Assumption (V2) where we treat the $\mathbf{Z}_{v,s}^j(m)$ as an independent sequence of random vectors in both the indices m and s . We omit the proof here, since it is the same proof as Lemma 2 of Lam and Feng (2018), but with $\alpha = 0$ on the volatility side of the proof.

In all the proofs below, if no ambiguity arises, we use n in a summation instead of $n(j)$ within a partition j to simplify notations since we assumed $n \asymp n(j)$ for each j anyway.

Lemma 3 *Let all the assumptions in Theorem 4 hold. Then with $p/n \rightarrow c \geq 0$, we have for a bandwidth Q defined in (3.8),*

$$\max_{\substack{i=1,\dots,p, j=1,\dots,L \\ l=1,\dots,Q-1, l'=1,\dots,Q-1 \\ l \neq l'}} \left| \sum_{s=1}^{n-Q+1} \frac{\mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_{s+l}} - \tilde{\mathbf{X}}_{v_{s+l-1}}) (\tilde{\mathbf{X}}_{v_{s+l'}} - \tilde{\mathbf{X}}_{v_{s+l'-1}})^T \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right| = O_P(p_f^{1/2} n^{-1/2}).$$

Proof of Lemma 3. We use the decomposition

$$\sum_{s=1}^{n-Q+1} \frac{\mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_{s+l}} - \tilde{\mathbf{X}}_{v_{s+l-1}}) (\tilde{\mathbf{X}}_{v_{s+l'}} - \tilde{\mathbf{X}}_{v_{s+l'-1}})^T \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} = \frac{P_{l,l'}^{(1)} + 2P_{l,l'}^{(2)} + P_{l,l'}^{(3)}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}},$$

where

$$\begin{aligned} P_{l,l'}^{(1)} &= \sum_{s=1}^{n-Q+1} (\mathbf{p}_{ij}^T \mathbf{A}(v_{s+l-1}, v_{s+l}) \mathbf{Z}_{d,s+l}^j + \mathbf{p}_{ij}^T \Sigma(v_{s+l-1}, v_{s+l})^{1/2} \mathbf{Z}_{v,s+l}^j) \\ &\quad \cdot (\mathbf{p}_{ij}^T \mathbf{A}(v_{s+l'-1}, v_{s+l'}) \mathbf{Z}_{d,s+l'}^j + \mathbf{p}_{ij}^T \Sigma(v_{s+l'-1}, v_{s+l'})^{1/2} \mathbf{Z}_{v,s+l'}^j)^T, \\ P_{l,l'}^{(2)} &= \sum_{s=1}^{n-Q+1} (\mathbf{p}_{ij}^T \mathbf{A}(v_{s+l-1}, v_{s+l}) \mathbf{Z}_{d,s+l}^j + \mathbf{p}_{ij}^T \Sigma(v_{s+l-1}, v_{s+l})^{1/2} \mathbf{Z}_{v,s+l}^j) (e(\mathbf{J}_{v_{s+l'}}) - e(\mathbf{J}_{v_{s+l'-1}}))^T \mathbf{p}_{ij}, \\ P_{l,l'}^{(3)} &= \sum_{s=1}^{n-Q+1} \mathbf{p}_{ij}^T (e(\mathbf{J}_{v_{s+l}}) - e(\mathbf{J}_{v_{s+l-1}})) (e(\mathbf{J}_{v_{s+l'}}) - e(\mathbf{J}_{v_{s+l'-1}}))^T \mathbf{p}_{ij}. \end{aligned}$$

Let us consider $P_{l,l'}^{(1)}$ first. We know that

$$\begin{aligned} P_{l,l'}^{(1)} &= \sum_{s=1}^{n-Q+1} (\mathbf{p}_{ij}^T \mathbf{A}(v_{s+l-1}, v_{s+l}) \mathbf{Z}_{d,s+l}^j \mathbf{Z}_{d,s+l'}^{jT} \mathbf{A}(v_{s+l'-1}, v_{s+l'})^T \mathbf{p}_{ij}) \\ &\quad + \sum_{s=1}^{n-Q+1} (\mathbf{p}_{ij}^T \mathbf{A}(v_{s+l-1}, v_{s+l}) \mathbf{Z}_{d,s+l}^j \mathbf{Z}_{v,s+l'}^{jT} \Sigma(v_{s+l'-1}, v_{s+l'})^{T/2} \mathbf{p}_{ij}) \\ &\quad + \sum_{s=1}^{n-Q+1} (\mathbf{p}_{ij}^T \Sigma(v_{s+l-1}, v_{s+l})^{1/2} \mathbf{Z}_{v,s+l}^j \mathbf{Z}_{d,s+l'}^{jT} \mathbf{A}(v_{s+l'-1}, v_{s+l'})^T \mathbf{p}_{ij}) \\ &\quad + \sum_{s=1}^{n-Q+1} (\mathbf{p}_{ij}^T \Sigma(v_{s+l-1}, v_{s+l})^{1/2} \mathbf{Z}_{v,s+l}^j \mathbf{Z}_{v,s+l'}^{jT} \Sigma(v_{s+l'-1}, v_{s+l'})^{T/2} \mathbf{p}_{ij}) \\ &= P_{l,l'}^{(1,1)} + P_{l,l'}^{(1,2)} + P_{l,l'}^{(1,3)} + P_{l,l'}^{(1,4)}. \end{aligned}$$

Consider $P_{l,l'}^{(1,1)}$. The proof is the same with non-random drift where then $p_f = p$ since $\mathbf{A}(\cdot, \cdot)$ contains just one non-zero column of vector. So here we only focus on proof with random drift.

We have

$$\begin{aligned} E_j |P_{l,l'}^{l,l'}| &\leq \sum_{s=1}^{n-Q+1} E_j^{1/2} (\mathbf{p}_{ij}^\top \mathbf{A}(v_{s+l-1}, v_{s+l}) \mathbf{Z}_{d,s+l}^j)^2 E_j^{1/2} (\mathbf{p}_{ij}^\top \mathbf{A}(v_{s+l'-1}, v_{s+l'}) \mathbf{Z}_{d,s+l'}^j)^2 \\ &= O(\|\mathbf{A}(v_{s+l-1}, v_{s+l})\| \|\mathbf{A}(v_{s+l'-1}, v_{s+l'})\|) = O(p_f n^{-1}). \end{aligned}$$

Also, using Lemma 2.7 of Bai and Silverstein (1998),

$$\begin{aligned} E_j |P_{l,l'}^{l,l'}|^2 &\leq \sum_{s=1}^{n-Q+1} E_j^{1/2} (\mathbf{p}_{ij}^\top \mathbf{A}(v_{s+l-1}, v_{s+l}) \mathbf{Z}_{d,s+l}^j)^4 E_j^{1/2} (\mathbf{p}_{ij}^\top \mathbf{A}(v_{s+l'-1}, v_{s+l'}) \mathbf{Z}_{d,s+l'}^j)^4 \\ &\quad + \sum_{s_1 \neq s_2} E_j ((\mathbf{p}_{ij}^\top \mathbf{A}(v_{s_1+l-1}, v_{s_1+l}) \mathbf{Z}_{d,s_1+l}^j \mathbf{p}_{ij}^\top \mathbf{A}(v_{s_2+l-1}, v_{s_2+l}) \mathbf{Z}_{d,s_2+l}^j) \\ &\quad \cdot (\mathbf{p}_{ij}^\top \mathbf{A}(v_{s_1+l'-1}, v_{s_1+l'}) \mathbf{Z}_{d,s_1+l'}^j \mathbf{p}_{ij}^\top \mathbf{A}(v_{s_2+l'-1}, v_{s_2+l'}) \mathbf{Z}_{d,s_2+l'}^j)) \\ &= O(n^2 \cdot \|\mathbf{A}(v_{s+l-1}, v_{s+l})\|^4) = O(p_f^2 n^{-2}). \end{aligned}$$

Hence we have

$$P_{l,l'}^{(1,1)} = O_P(p_f n^{-1}).$$

Same technique shows that

$$P_{l,l'}^{(1,2)}, P_{l,l'}^{(1,3)} = O_P(p_f n^{-1/2}).$$

Now consider $P_{l,l'}^{(1,4)}$. By Assumption (V1), for $\ell' < \ell$, defining

$$R_{\ell,\ell'} = \mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(v_{\ell-1}, v_\ell)^{1/2} \mathbf{Z}_{v,\ell}^j \mathbf{Z}_{v,\ell'}^{j\top} \boldsymbol{\Sigma}(v_{\ell'-1}, v_{\ell'})^{1/2} \mathbf{p}_{ij},$$

we can use the independence of the $\mathbf{Z}_{v,s}^j$'s to see that $E(R_{\ell,\ell'} | \mathcal{F}_{-j} \cup \mathcal{F}_{\ell-1}^j) = 0$. Hence using the Burkholder's inequality, we have

$$E_j (P_{l,l'}^{(1,4)})^2 \leq C \sum_{s=1}^{n-Q+1} E_j (R_{s+l,s+l'}^2).$$

where C is a generic constant. Using Lemma 2.7 of Bai and Silverstein (1998),

$$\begin{aligned} E(R_{\ell,\ell'}^2) &\leq 2E(R_{\ell,\ell}^2 + R_{\ell',\ell'}^2), \quad \text{with} \\ E_j(R_{\ell,\ell}^2) &= O(\|\boldsymbol{\Sigma}(v_{\ell-1}, v_\ell)\|^2) = O(p_f^2 n^{-2}). \end{aligned}$$

Hence we can conclude that $P_{l,l'}^{(1,4)} = O_P(p_f n^{-1/2})$, and hence

$$\begin{aligned} P_{l,l'}^{(1)} / \mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij} &= (P_{l,l'}^{(1,1)} + P_{l,l'}^{(1,2)} + P_{l,l'}^{(1,3)} + P_{l,l'}^{(1,4)}) / \mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij} \\ &= O_P(p_f^{1/2} n^{-1/2}). \end{aligned}$$

Using the jumps removal rate in Fan and Wang (2007), and Assumption (W2) and (W3) that there are only finite number of jumps for each stock and cojumps at each time point, we have

$$P_{l,l'}^{(3)} / \mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij} = O_P(n^{-1/2}).$$

For $P_{l,l'}^{(2)}$, similar to the treatment above,

$$P_{l,l'}^{(2)} = O_P(p_f^{1/2} n^{-5/4} + p_f^{1/2} n^{-3/4}),$$

which is dominated by $P_{l,l'}^{(1)}$. Hence we have

$$\max_{\substack{i=1,\dots,p, j=1,\dots,L \\ l=1,\dots,Q-1, l'=1,\dots,Q-1 \\ l \neq l'}} \left| \sum_{s=1}^{n-Q+1} \frac{\mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_{s+l}} - \tilde{\mathbf{X}}_{v_{s+l-1}}) (\tilde{\mathbf{X}}_{v_{s+l'}} - \tilde{\mathbf{X}}_{v_{s+l'-1}})^T \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right| = O_P(p_f^{1/2} n^{-1/2}). \quad \square$$

Lemma 4 *Let all the assumptions in Theorem 4 hold. Then with $p/n \rightarrow c \geq 0$,*

$$\max_{\substack{i=1,\dots,p \\ j=1,\dots,L}} \left| \sum_{s=1}^n \frac{\mathbf{p}_{ij}^T (\mathbf{E}(s) \mathbf{E}(s)^T) \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right| = O_P(n).$$

Proof of Lemma 4. By the definition of $\mathbf{E}(s)$, we have

$$\begin{aligned} \sum_{s=1}^n \frac{\mathbf{p}_{ij}^T (\mathbf{E}(s) \mathbf{E}(s)^T) \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} &= \sum_{\ell=1}^3 E_\ell + 2 \sum_{\ell=4}^6 E_\ell, \text{ where} \\ E_1 &= \sum_{s=1}^n \frac{(\mathbf{p}_{ij}^T (\boldsymbol{\epsilon}(s)))^2}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ E_2 &= \sum_{s=1}^n \frac{(\mathbf{p}_{ij}^T (\mathbf{X}(s) - \mathbf{X}_{v_s}))^2}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ E_3 &= \sum_{s=1}^n \frac{(\mathbf{p}_{ij}^T (e(\mathbf{J}(s)) - e(\mathbf{J}_{v_s})))^2}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ E_4 &= \sum_{s=1}^n \frac{\mathbf{p}_{ij}^T (\boldsymbol{\epsilon}(s)) (\mathbf{X}(s) - \mathbf{X}_{v_s})^T \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ E_5 &= \sum_{s=1}^n \frac{\mathbf{p}_{ij}^T (\boldsymbol{\epsilon}(s)) (e(\mathbf{J}(s)) - e(\mathbf{J}_{v_s}))^T \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ E_6 &= \sum_{s=1}^n \frac{\mathbf{p}_{ij}^T (\mathbf{X}(s) - \mathbf{X}_{v_s}) (e(\mathbf{J}(s)) - e(\mathbf{J}_{v_s}))^T \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}. \end{aligned}$$

By Lemma 2, and Assumption (W2) and (W3) together with the rate of jumps removal in Fan and Wang (2007), we have

$$E_2 = O_P(1), \quad E_3 = O_P(n^{-1/2}).$$

For E_1 , consider $E_1 = E_{1,1} + E_{1,2}$, where

$$\begin{aligned} E_{1,1} &= \frac{\sum_{s=1}^n ((\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s))^2 - \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon,s}^j \mathbf{p}_{ij})}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ E_{1,2} &= \frac{\sum_{s=1}^n \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon,s}^j \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}. \end{aligned} \tag{S.3}$$

For $E_{1,2}$, Assumption (E1) implies that

$$E_{1,2} = O_P(n).$$

For $E_{1,1}$, let $g_s^{ij} = (\mathbf{p}_{ij}^\top \boldsymbol{\epsilon}(s))^2 - \mathbf{p}_{ij}^\top \boldsymbol{\Sigma}_{\epsilon,s}^j \mathbf{p}_{ij}$. Using Lemma 2.7 of Bai and Silverstein (1998) under Assumption (E1) to (E3), we have

$$\begin{aligned} E(E_{1,1}^2 | \mathcal{F}_{-j} \cup \{\boldsymbol{\Sigma}_{\epsilon,u}, u \in [0, 1]\}) &= (\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{-2} \left\{ \sum_{s=1}^n E((g_s^{ij})^2 | \mathcal{F}_{-j} \cup \{\boldsymbol{\Sigma}_{\epsilon,u}, u \in [0, 1]\}) \right. \\ &\quad \left. + \sum_{s_1 \neq s_2} E(g_{s_1}^{ij} g_{s_2}^{ij} | \mathcal{F}_{-j} \cup \{\boldsymbol{\Sigma}_{\epsilon,u}, u \in [0, 1]\}) \right\} \\ &= O(n \cdot 1 + n^2 \cdot n^{-1} \cdot 1) = O(n). \end{aligned}$$

Hence $E_{1,1} = O_P(n^{1/2})$, and so

$$E_1 = O_P(n).$$

Finally, a simple C - r inequality shows that either E_1 , E_2 or E_3 dominates the order of E_4 to E_6 , and hence

$$\sum_{s=1}^n \frac{\mathbf{p}_{ij}^\top (\mathbf{E}(s) \mathbf{E}(s)^\top) \mathbf{p}_{ij}}{\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} = O_P(n). \quad \square$$

Lemma 5 *Let all the assumptions in Theorem 4 hold. Then with $p/n \rightarrow c \geq 0$, for $0 < l < n$,*

$$\sum_{l'=0}^{l-1} \sum_{s=1}^{n-l} \frac{\mathbf{p}_{ij}^\top \mathbf{E}(s+l) \mathbf{E}(s+l')^\top \mathbf{p}_{ij}}{\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} = O_P(l^{1/2} n^{1/2}).$$

Proof of Lemma 5. We can decompose $\sum_{l'=0}^{l-1} \sum_{s=1}^{n-l} \frac{\mathbf{p}_{ij}^\top \mathbf{E}(s+l) \mathbf{E}(s+l')^\top \mathbf{p}_{ij}}{\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} = \sum_{j=1}^9 L_j$, where

$$\begin{aligned} L_1 &= \sum_{s=1}^{n-l} \frac{\mathbf{p}_{ij}^\top \boldsymbol{\epsilon}(s+l) \sum_{l'=0}^{l-1} \boldsymbol{\epsilon}(s+l')^\top \mathbf{p}_{ij}}{\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ L_2 &= \sum_{s=1}^{n-l} \frac{\mathbf{p}_{ij}^\top (\mathbf{X}(s+l) - \mathbf{X}_{v_{s+l}}) \sum_{l'=0}^{l-1} (\mathbf{X}(s+l') - \mathbf{X}_{v_{s+l'}})^\top \mathbf{p}_{ij}}{\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ L_3 &= \sum_{s=1}^{n-l} \frac{\mathbf{p}_{ij}^\top (e(\mathbf{J}(s+l)) - e(\mathbf{J}_{v_{s+l}})) \sum_{l'=0}^{l-1} (e(\mathbf{J}(s+l')) - e(\mathbf{J}_{v_{s+l'}}))^\top \mathbf{p}_{ij}}{\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ L_4 &= \sum_{s=1}^{n-l} \frac{\mathbf{p}_{ij}^\top \boldsymbol{\epsilon}(s+l) \sum_{l'=0}^{l-1} (\mathbf{X}(s+l') - \mathbf{X}_{v_{s+l'}})^\top \mathbf{p}_{ij}}{\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ L_5 &= \sum_{s=1}^{n-l} \frac{\mathbf{p}_{ij}^\top \boldsymbol{\epsilon}(s+l) \sum_{l'=0}^{l-1} (e(\mathbf{J}(s+l')) - e(\mathbf{J}_{v_{s+l'}}))^\top \mathbf{p}_{ij}}{\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ L_6 &= \sum_{s=1}^{n-l} \frac{\mathbf{p}_{ij}^\top (\mathbf{X}(s+l) - \mathbf{X}_{v_{s+l}}) \sum_{l'=0}^{l-1} (e(\mathbf{J}(s+l')) - e(\mathbf{J}_{v_{s+l'}}))^\top \mathbf{p}_{ij}}{\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ L_7 &= \sum_{s=1}^{n-l} \frac{\mathbf{p}_{ij}^\top \sum_{l'=0}^{l-1} \boldsymbol{\epsilon}(s+l') (\mathbf{X}(s+l) - \mathbf{X}_{v_{s+l}})^\top \mathbf{p}_{ij}}{\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ L_8 &= \sum_{s=1}^{n-l} \frac{\mathbf{p}_{ij}^\top \sum_{l'=0}^{l-1} \boldsymbol{\epsilon}(s+l') (e(\mathbf{J}(s+l)) - e(\mathbf{J}_{v_{s+l}}))^\top \mathbf{p}_{ij}}{\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ L_9 &= \sum_{s=1}^{n-l} \frac{\mathbf{p}_{ij}^\top \sum_{l'=0}^{l-1} (\mathbf{X}(s+l') - \mathbf{X}_{v_{s+l'}}) (e(\mathbf{J}(s+l)) - e(\mathbf{J}_{v_{s+l}}))^\top \mathbf{p}_{ij}}{\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}. \end{aligned}$$

Consider L_1 first. Denoting $E_j(\cdot) = E(\cdot | \mathcal{F}_{-j})$, and $\text{cov}_j(\cdot, \cdot)$ the corresponding covariance operator given $\mathcal{F}_{-j}$, we have

$$\begin{aligned} E_j(L_1) &= \sum_{s=1}^{n-l} O(1) \cdot \text{cov}_j \left(\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s+l), \mathbf{p}_{ij}^T \sum_{l'=0}^{l-1} \boldsymbol{\epsilon}(s+l') \right) \\ &\leq \sum_{s=1}^{n-l} O(1) \cdot O(n^{-1}) (\mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s+l}^j \mathbf{p}_{ij})^{1/2} O(l + l^2 n^{-1})^{1/2} = O(l^{1/2}), \end{aligned}$$

where we used Assumption (E3) on the weak correlation (at order n^{-1}) between functions of $\boldsymbol{\epsilon}(s_1)$ and $\boldsymbol{\epsilon}(s_2)$ when $s_1 \neq s_2$, and the boundedness of $\|\boldsymbol{\Sigma}_{\epsilon, s}^j\|$ in Assumption (E1). Also,

$$\begin{aligned} E_j(L_1^2) &= \sum_{s_1, s_2=1}^{n-l} O(1) \cdot E_j \left(\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s_1+l) \mathbf{p}_{ij}^T \sum_{l'=0}^{l-1} \boldsymbol{\epsilon}(s_1+l') \mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s_2+l) \mathbf{p}_{ij}^T \sum_{l'=0}^{l-1} \boldsymbol{\epsilon}(s_2+l') \right) \\ &= O(1) \cdot O(nl + n^2 \cdot n^{-1} \cdot l) = O(nl), \end{aligned}$$

so that

$$L_1 = O_P(n^{1/2} l^{1/2}).$$

To find the rate of L_2 , define

$$\begin{aligned} A_d^{ij}(s) &= \sum_{m=1}^{p-b_s-1} \frac{\mathbf{p}_{ij}^T \mathbf{D}_m^s \mathbf{A}(t_s^{j_m}, t_s^{j_{m+1}}) \mathbf{Z}_{d,s}^j(m)}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}}, \\ A_v^{ij}(s) &= \sum_{m=1}^{p-b_s-1} \frac{\mathbf{p}_{ij}^T \mathbf{D}_m^s \boldsymbol{\Sigma}(t_s^{j_m}, t_s^{j_{m+1}})^{1/2} \mathbf{Z}_{v,s}^j(m)}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}}. \end{aligned} \tag{S.4}$$

Then we can decompose $L_2 = L_{d,d} + L_{d,v} + L_{v,d} + L_{v,v}$, where

$$L_{d,v} = \sum_{s=1}^{n-l} A_d^{ij}(s+l) \sum_{l'=0}^{l-1} A_v^{ij}(s+l'), \quad L_{v,d} = \sum_{s=1}^{n-l} A_v^{ij}(s+l) \sum_{l'=0}^{l-1} A_d^{ij}(s+l'),$$

and the two other terms are defined similarly. Going through the proof of Lemma 2 in Lam and Feng (2018), we have

$$A_v^{ij}(s) = O_p(n^{-1/2}), \quad A_d^{ij}(s) = \begin{cases} O_P(p_f^{1/2} n^{-1}), & \text{non-random drift;} \\ O_P(p_f^{1/2} p^{\alpha-1/2} n^{-1}), & \text{random drift.} \end{cases}$$

Hence for non-random drift, using Burkholder's inequality on all except $L_{d,d}$,

$$L_{d,d} = O_P(p_f l n^{-1}), \quad L_{d,v}, L_{v,d} = O_P(p_f^{1/2} l n^{1/2} \cdot n^{-1/2} \cdot n^{-1}) = O_P(p_f^{1/2} l n^{-1}), \quad L_{v,v} = O_P(l^{1/2} n^{-1/2}),$$

implying that

$$L_2 = O_P(p_f l n^{-1} + l^{1/2} n^{-1/2}),$$

which in fact is true for the case of random drift as well.

Using Assumption (W1) to (W3), we have

$$\begin{aligned} L_3 &= O_p(l n^{-1/2}), \quad L_5, L_8 = O_P(l n^{-1/4}), \\ L_6, L_9 &= O_P(p_f^{1/2} \cdot l n^{-1/2} \cdot n^{-1/4}) = O_P(p_f^{1/2} l n^{-3/4}). \end{aligned}$$

Finally, using Assumption (E3) and the fact that $\{A_v^{ij}(s)\}$ is a sequence of independent variables in the index s ,

$$L_4, L_7 = O_P(n^{1/2} \cdot (p_f^{1/2} l n^{-1} + l^{1/2} n^{-1/2})) = O_P(p_f^{1/2} l n^{-1/2} + l^{1/2}).$$

Hence we have the result as stated, since $l^{1/2} n^{1/2}$ dominate all other term no matter $p_f = 1$ or $p_f = p \asymp n$. $\square$

Proof of Theorem 1

For NER-MSRVM in (3.3), $i = 1, \dots, p$, $j = 1, \dots, L$ with $\mathbf{P}_{-j} = (\mathbf{p}_{1j}, \dots, \mathbf{p}_{pj})$, we can decompose

$$\begin{aligned} \mathbf{p}_{ij}^T \tilde{\Sigma}(\tau_{j-1}, \tau_j)^M \mathbf{p}_{ij} &= \mathbf{p}_{ij}^T \sum_{m=1}^M a_m [\tilde{\mathbf{Y}}, \tilde{\mathbf{Y}}^T]_j^{(K_m)} \mathbf{p}_{ij} + \mathbf{p}_{ij}^T \zeta \left([\tilde{\mathbf{Y}}, \tilde{\mathbf{Y}}^T]_j^{(K_1)} - [\tilde{\mathbf{Y}}, \tilde{\mathbf{Y}}^T]_j^{(K_M)} \right) \mathbf{p}_{ij} \\ &= I_1 + 2I_2 + I_3, \end{aligned}$$

where $\tilde{\Sigma}(\tau_{j-1}, \tau_j)^M$ is the MSRVM in (3.4) constructed using jumps-removed data, and

$$\begin{aligned} I_1 &= \mathbf{p}_{ij}^T \sum_{m=1}^M a_m [\tilde{\mathbf{X}}_v, \tilde{\mathbf{X}}_v^T]_j^{(K_m)} \mathbf{p}_{ij} + \mathbf{p}_{ij}^T \zeta \left([\tilde{\mathbf{X}}_v, \tilde{\mathbf{X}}_v^T]_j^{(K_1)} - [\tilde{\mathbf{X}}_v, \tilde{\mathbf{X}}_v^T]_j^{(K_M)} \right) \mathbf{p}_{ij} \\ &= \sum_{m=1}^M a_m \mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \tilde{\mathbf{X}}_v^T]_j^{(K_m)} \mathbf{p}_{ij} + \zeta \mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \tilde{\mathbf{X}}_v^T]_j^{(K_1)} \mathbf{p}_{ij} - \zeta \mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \tilde{\mathbf{X}}_v^T]_j^{(K_M)} \mathbf{p}_{ij}, \\ I_2 &= \sum_{m=1}^M a_m \mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \mathbf{E}^T]_j^{(K_m)} \mathbf{p}_{ij} + \zeta \mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \mathbf{E}^T]_j^{(K_1)} \mathbf{p}_{ij} - \zeta \mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \mathbf{E}^T]_j^{(K_M)} \mathbf{p}_{ij}, \\ I_3 &= \sum_{m=1}^M a_m \mathbf{p}_{ij}^T [\mathbf{E}, \mathbf{E}^T]_j^{(K_m)} \mathbf{p}_{ij} + \zeta \mathbf{p}_{ij}^T [\mathbf{E}, \mathbf{E}^T]_j^{(K_1)} \mathbf{p}_{ij} - \zeta \mathbf{p}_{ij}^T [\mathbf{E}, \mathbf{E}^T]_j^{(K_M)} \mathbf{p}_{ij}. \end{aligned}$$

Going through the proof of Lemma 3 in Lam and Feng (2018), we have

$$\max_{\substack{i=1, \dots, p \\ j=1, \dots, L}} \left| \frac{\mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \mathbf{E}^T]_j^{(K_m)} \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right| = O_P(K_m^{-1/2} + n^{-1/4}). \quad (\text{S.5})$$

By definition, $K_m = N + m$, with $N \asymp n^{2/3}$ and $M \asymp n^{1/2}$. Hence it is easy to see that all K_m has the same order at $n^{2/3}$ for $m = 1, \dots, M$. Also, since $\sum_{m=1}^M a_m = 1$ and $\zeta \asymp n^{-1/6}$, it is easy to show that

$$\begin{aligned} \left| \sum_{m=1}^M a_m \frac{\mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \tilde{\mathbf{X}}_v^T]_j^{(K_m)} \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1 \right| &= O_P(K_m^{1/2} n^{-1/2} + p_f^{1/2} n^{-1/2}) \\ &= O_P(n^{-1/6} + p_f^{1/2} n^{-1/2}), \\ \zeta \cdot \left(\left| \frac{\mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \tilde{\mathbf{X}}_v^T]_j^{(K_1)} \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right| + \left| \frac{\mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \tilde{\mathbf{X}}_v^T]_j^{(K_M)} \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right| \right) &= O_p(n^{-1/6} \cdot (K_m^{1/2} n^{-1/2} + p_f^{1/2} n^{-1/2} + 1)) \\ &= O_P(n^{-1/6}). \end{aligned}$$

where both results above used Lemma 1. Since the above bounds are independent of the indices i and j , we have established that

$$\max_{\substack{i=1,\dots,p \\ j=1,\dots,L}} \left| \frac{I_1}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1 \right| = O_P(n^{-1/6} + p_f^{1/2} n^{-1/2}). \quad (\text{S.6})$$

Similarly, to find the rate of I_2 , using (S.5),

$$\begin{aligned} \sum_{m=1}^M a_m \frac{\mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \mathbf{E}^T]_j^{(K_m)} \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} &= O_P(K_m^{-1/2} + n^{-1/4}) = O_P(n^{-1/4}), \\ \zeta \left(\frac{\mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \mathbf{E}^T]_j^{(K_1)} \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - \frac{\mathbf{p}_{ij}^T [\tilde{\mathbf{X}}_v, \mathbf{E}^T]_j^{(K_M)} \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right) &= O_P(n^{-1/6} \cdot (K_m^{-1/2} + n^{-1/4})) = O_P(n^{-5/12}). \end{aligned}$$

Since the above two results are free of all indices i and j ,

$$\max_{\substack{i=1,\dots,p \\ j=1,\dots,L}} \left| \frac{I_2}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right| = O_P(n^{-1/4}). \quad (\text{S.7})$$

To find the rate for I_3 , consider the decomposition

$$\begin{aligned} \frac{\mathbf{p}_{ij}^T [\mathbf{E}, \mathbf{E}^T]_j^{(K_m)} \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} &= \sum_{\ell=1}^3 I_{3,\ell} + 2 \sum_{\ell=1}^3 I_{3,\ell}, \text{ where} \\ I_{3,1}(m) &= \frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} \frac{(\mathbf{p}_{ij}^T (\boldsymbol{\epsilon}(s) - \boldsymbol{\epsilon}(s - K_m)))^2}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ I_{3,2} &= \frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} \frac{(\mathbf{p}_{ij}^T (\mathbf{X}(s) - \mathbf{X}_{v_s} + \mathbf{X}_{v_{s-K_m}} - \mathbf{X}(s - K_m)))^2}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ I_{3,3} &= \frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} \frac{(\mathbf{p}_{ij}^T (e(\mathbf{J}(s)) - e(\mathbf{J}_{v_s}) - e(\mathbf{J}(s - K_m)) + e(\mathbf{J}_{v_{s-K_m}})))^2}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ I_{3,4} &= \frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} \frac{\mathbf{p}_{ij}^T (\boldsymbol{\epsilon}(s) - \boldsymbol{\epsilon}(s - K_m)) (\mathbf{X}(s) - \mathbf{X}_{v_s} + \mathbf{X}_{v_{s-K_m}} - \mathbf{X}(s - K_m))^T \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ I_{3,5} &= \frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} \frac{\mathbf{p}_{ij}^T (\boldsymbol{\epsilon}(s) - \boldsymbol{\epsilon}(s - K_m)) (e(\mathbf{J}(s)) - e(\mathbf{J}_{v_s}) - e(\mathbf{J}(s - K_m)) + e(\mathbf{J}_{v_{s-K_m}}))^T \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ I_{3,6} &= \frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} \frac{\mathbf{p}_{ij}^T (\mathbf{X}(s) - \mathbf{X}_{v_s} + \mathbf{X}_{v_{s-K_m}} - \mathbf{X}(s - K_m))}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \\ &\quad \cdot (e(\mathbf{J}(s)) - e(\mathbf{J}_{v_s}) - e(\mathbf{J}(s - K_m)) + e(\mathbf{J}_{v_{s-K_m}}))^T \mathbf{p}_{ij}. \end{aligned}$$

We consider $I_{3,2}$ first, which by Lemma 2 has

$$I_{3,2} = O_P(n K_m^{-1} \cdot n^{-1}) = O_P(K_m^{-1}) = O_P(n^{-2/3}).$$

Using Assumption (W1) to (W3) and the rate $n^{-1/4}$ of jumps removal in Fan and Wang (2007), we have

$$\begin{aligned} I_{3,3} &= O_P(n^{-1/2}), \quad I_{3,5} = O_P(n^{-1/4}), \\ I_{3,6} &= O_P(n^{-1/2} n^{-1/4}) = O_P(n^{-3/4}). \end{aligned}$$

Using Assumption (E3), we have

$$I_{3,4} = O_P(K_m^{-2} \cdot n \cdot n^{-1}) = O_P(n^{-2/3}).$$

Instead of considering the rate of $I_{3,1}(m)$, we consider

$$\begin{aligned} & \mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij} \left(\sum_{m=1}^M a_m I_{3,1}(K_m) + \zeta(I_{3,1}(K_1) - I_{3,1}(K_M)) \right) \\ &= \sum_{m=1}^M a_m \frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} (\mathbf{p}_{ij}^T (\boldsymbol{\epsilon}(s) - \boldsymbol{\epsilon}(s - K_m)))^2 \\ & \quad + \zeta \left(\frac{1}{K_1} \sum_{s, s-K_1 \in S^j(K_1)} (\mathbf{p}_{ij}^T (\boldsymbol{\epsilon}(s) - \boldsymbol{\epsilon}(s - K_1)))^2 - \frac{1}{K_M} \sum_{s, s-K_M \in S^j(K_M)} (\mathbf{p}_{ij}^T (\boldsymbol{\epsilon}(s) - \boldsymbol{\epsilon}(s - K_M)))^2 \right) \\ &= J_1 - 2J_2 + J_3, \end{aligned} \tag{S.8}$$

where

$$\begin{aligned} J_1 &= \sum_{m=1}^M a_m \frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} (\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s))^2 \\ & \quad + \zeta \left(\frac{1}{K_1} \sum_{s, s-K_1 \in S^j(K_1)} (\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s))^2 - \frac{1}{K_M} \sum_{s, s-K_M \in S^j(K_M)} (\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s))^2 \right), \\ J_2 &= \sum_{m=1}^M a_m \frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} (\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s) \boldsymbol{\epsilon}(s - K_m)^T \mathbf{p}_{ij}) \\ & \quad + \zeta \left(\frac{1}{K_1} \sum_{s, s-K_1 \in S^j(K_1)} (\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s) \boldsymbol{\epsilon}(s - K_1)^T \mathbf{p}_{ij}) - \frac{1}{K_M} \sum_{s, s-K_M \in S^j(K_M)} (\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s) \boldsymbol{\epsilon}(s - K_M)^T \mathbf{p}_{ij}) \right), \\ J_3 &= \sum_{m=1}^M a_m \frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} (\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s - K_m))^2 \\ & \quad + \zeta \left(\frac{1}{K_1} \sum_{s, s-K_1 \in S^j(K_1)} (\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s - K_1))^2 - \frac{1}{K_M} \sum_{s, s-K_M \in S^j(K_M)} (\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s - K_M))^2 \right). \end{aligned}$$

Writing $g_{m,s}^{ij} = \mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s) \boldsymbol{\epsilon}(s - K_m)^T \mathbf{p}_{ij}$, by Lemma 2.7 of Bai and Silverstein (1998),

$$\begin{aligned} E \left\{ \left(\frac{1}{K_m} \sum_{s, s-K_m \in S^j(m)} g_{m,s}^{ij} \right)^2 \middle| \mathcal{F}_{ij} \right\} &= O(K_m^{-2} n \cdot 1 + n^{-1} \cdot K_m^{-2} n^2 \cdot 1) = O(K_m^{-2} n), \text{ hence} \\ \frac{1}{K_m} \sum_{s, s-K_m \in S^j(m)} g_{m,s}^{ij} &= O_P(K_m^{-1} n^{1/2}) = O_P(n^{-1/6}), \end{aligned}$$

which implies that

$$J_2 = O_P(n^{-1/6} + n^{-1/3}) = O_P(n^{-1/6}).$$

We can further decompose $J_1 = J_{11} + J_{12} + J_{13}$, where

$$\begin{aligned}
J_{11} &= \sum_{m=1}^M a_m \frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} ((\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s))^2 - \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij}), \\
J_{12} &= \zeta \left(\frac{1}{K_1} \sum_{s, s-K_1 \in S^j(K_1)} ((\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s))^2 - \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij}) - \frac{1}{K_M} \sum_{s, s-K_M \in S^j(K_M)} ((\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s))^2 - \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij}) \right), \\
J_{13} &= \sum_{m=1}^M a_m \frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij} \\
&\quad + \zeta \left(\frac{1}{K_1} \sum_{s, s-K_1 \in S^j(K_1)} \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij} - \frac{1}{K_M} \sum_{s, s-K_M \in S^j(K_M)} \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij} \right).
\end{aligned}$$

Now define $g_s^{ij} = \mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s) - \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij}$. Using Lemma 2.7 of Bai and Silverstein (1998) under Assumption (E1) to (E3), we have

$$\begin{aligned}
E_j \left(\frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} (g_s^{ij})^2 | \{ \boldsymbol{\Sigma}_{\epsilon, u}, u \in [0, 1] \} \right) &= K_m^{-2} \sum_{s, s-K_m \in S^j(K)} E_j((g_s^{ij})^2 | \{ \boldsymbol{\Sigma}_{\epsilon, u}, u \in [0, 1] \}) \\
&\quad + K_m^{-2} \sum_{s_1 \neq s_2} E_j(g_{s_1}^{ij} g_{s_2}^{ij} | \{ \boldsymbol{\Sigma}_{\epsilon, u}, u \in [0, 1] \}) \\
&= O(K_m^{-2} n \cdot 1 + K_m^{-2} n^2 \cdot n^{-1} \cdot 1) = O_P(n^{-1/3}).
\end{aligned}$$

The above implies that

$$J_{11} = O_P(n^{-1/6}) = J_{12}.$$

As for the rate of J_{13} , we can do a further decomposition $J_{13} = J_{13}^{(1)} + J_{13}^{(2)} + J_{13}^{(3)}$, where $c_m = \frac{n-K_m+1}{K_m}$ and

$$\begin{aligned}
J_{13}^{(1)} &= \sum_{m=1}^M a_m \left(\frac{1}{K_m} \sum_{s, s-K_m \in S^j(K_m)} \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij} - c_m \mathbf{p}_{ij}^T E(\boldsymbol{\Sigma}_{\epsilon, s}^j) \mathbf{p}_{ij} \right), \\
J_{13}^{(2)} &= \zeta \left(\left(\frac{1}{K_1} \sum_{s, s-K_1 \in S^j(K_1)} \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij} - c_1 \mathbf{p}_{ij}^T E(\boldsymbol{\Sigma}_{\epsilon, s}^j) \mathbf{p}_{ij} \right) \right. \\
&\quad \left. - \left(\frac{1}{K_M} \sum_{s, s-K_M \in S^j(K_M)} \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij} - c_M \mathbf{p}_{ij}^T E(\boldsymbol{\Sigma}_{\epsilon, s}^j) \mathbf{p}_{ij} \right) \right), \\
J_{13}^{(3)} &= \left(\sum_{m=1}^M a_m c_m + \zeta(c_1 - c_M) \right) \mathbf{p}_{ij}^T E(\boldsymbol{\Sigma}_{\epsilon, s}^j) \mathbf{p}_{ij}.
\end{aligned}$$

Parallel to the argument used for J_{11} , under Assumption (E3), we have

$$J_{13}^{(1)} = O_P(K_m^{-2} n + K_m^{-2} n^2 \cdot n^{-1})^{1/2} = O_P(n^{-1/6}).$$

Similarly,

$$J_{13}^{(2)} = O_P(n^{-1/6} \cdot n^{-1/6}) = O_P(n^{-1/3}).$$

By the same technique used in the proof of Theorem 1 in Tao et al. (2013), and the definition of a_m , ζ in 3.3 and c_m shown above, we have $J_{13}^{(3)} = 0$. Hence

$$J_{13} = O_P(n^{-1/6}).$$

This implies that

$$J_1 = O_P(n^{-1/6}) = J_3.$$

Combining these rates, we have

$$\sum_{m=1}^M a_m I_{3,1}(K_m) + \zeta(I_{3,1}(K_1) - I_{3,1}(K_M)) = O_P(n^{-1/6}).$$

Hence this, together with the rates for $I_{3,2}$ to $I_{3,6}$, allow us to conclude that

$$\max_{\substack{i=1,\dots,p \\ j=1,\dots,L}} \left| \frac{I_3}{\mathbf{P}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{P}_{ij}} \right| = O_P(n^{-1/6}). \quad (\text{S.9})$$

Combining (S.6), (S.7) and (S.9), we thus have

$$\max_{\substack{i=1,\dots,p \\ j=1,\dots,L}} \left| \frac{\mathbf{P}_{ij}^T \widetilde{\boldsymbol{\Sigma}}(\tau_{j-1}, \tau_j)^M \mathbf{P}_{ij}}{\mathbf{P}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{P}_{ij}} - 1 \right| = O_P(n^{-1/6} + p_f^{1/2} n^{-1/2}).$$

For $\widehat{\boldsymbol{\Sigma}}(0, 1)^M$, we have

$$\begin{aligned} \left\| \widehat{\boldsymbol{\Sigma}}(0, 1)^M \boldsymbol{\Sigma}_{\text{Ideal}}(0, 1)^{-1} - \mathbf{I}_p \right\| &= \left\| \sum_{j=1}^L (\widehat{\boldsymbol{\Sigma}}(\tau_{j-1}, \tau_j)^M \boldsymbol{\Sigma}_{\text{Ideal}}(\tau_{j-1}, \tau_j)^{-1} - \mathbf{I}_p) \boldsymbol{\Sigma}_{\text{Ideal}}(\tau_{j-1}, \tau_j) \boldsymbol{\Sigma}_{\text{Ideal}}(0, 1)^{-1} \right\| \\ &\leq \sum_{j=1}^L \left\| \widehat{\boldsymbol{\Sigma}}(\tau_{j-1}, \tau_j)^M \boldsymbol{\Sigma}_{\text{Ideal}}(\tau_{j-1}, \tau_j)^{-1} - \mathbf{I}_p \right\| \\ &\quad \cdot \left\| \text{diag}(\mathbf{P}_{-j}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{P}_{-j}) \cdot \left(\text{diag}(\mathbf{P}_{-j}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{P}_{-j}) + \sum_{i \neq j} \mathbf{P}_{-j}^T \boldsymbol{\Sigma}_{\text{Ideal}}(\tau_{i-1}, \tau_i) \mathbf{P}_{-j} \right)^{-1} \right\| \\ &= O_P \left((n^{-1/6} + p_f^{1/2} n^{-1/2}) \max_{j=1,\dots,L} \left\| \left(\mathbf{I}_p + \sum_{i \neq j} \mathbf{P}_{-j}^T \boldsymbol{\Sigma}_{\text{Ideal}}(\tau_{i-1}, \tau_i) \mathbf{P}_{-j} \text{diag}^{-1}(\mathbf{P}_{-j}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{P}_{-j}) \right)^{-1} \right\| \right) \\ &= O_P(n^{-1/6} + p_f^{1/2} n^{-1/2}). \quad \square \end{aligned}$$

Proof of the Theorem 2

In this proof we drop the subscripts j in $\boldsymbol{\gamma}_j^{(h)}(\widetilde{\mathbf{Y}}_j^{(J)})$ since we are working within partition j already. Define, for a time series of vectors $\mathbf{Y}(\cdot)$,

$$\begin{aligned} \mathbf{Y}^{(J)}(1) &= J^{-1} \sum_{l=0}^{J-1} \mathbf{Y}(l), \quad \mathbf{Y}^{(J)}(n-2J+1) = J^{-1} \sum_{l=n-J+1}^n \mathbf{Y}(l), \\ \mathbf{Y}^{(J)}(s) &= \widetilde{\mathbf{Y}}(s+J-1), \quad s = 2, \dots, n-2J. \end{aligned}$$

To extract the jittering effects for analysis, define

$$\begin{aligned} \widetilde{\mathbf{Y}}^{*(J)}(1) &= \widetilde{\mathbf{X}}(J), \quad \widetilde{\mathbf{Y}}^{*(J)}(n-2J+1) = \widetilde{\mathbf{X}}(n-J), \\ \widetilde{\mathbf{Y}}^{*(J)}(s) &= \widetilde{\mathbf{Y}}^{(J)}(s) = \widetilde{\mathbf{Y}}(s+J-1), \quad s = 2, \dots, n-2J. \end{aligned}$$

Then we have

$$\mathbf{p}_{ij}^T \mathbf{K}(\widetilde{\mathbf{Y}}^{(J)}) \mathbf{p}_{ij} - \mathbf{p}_{ij}^T \mathbf{K}(\widetilde{\mathbf{Y}}^{*(J)}) \mathbf{p}_{ij} = \mathbf{p}_{ij}^T \mathbf{E}_J^{(0)} \mathbf{p}_{ij} + 2 \sum_{h=1}^{n-2J} k_H(h) \mathbf{p}_{ij}^T \mathbf{E}_J^{(h)} \mathbf{p}_{ij},$$

where $k_H(h) = k((h-1)/H)$ or $k(h/H)$, and $\mathbf{E}_J^{(h)} = \mathbf{E}_{J,1}^{(h)} + \mathbf{E}_{J,2}^{(h)}$, where for $h > 0$,

$$\begin{aligned}\mathbf{E}_{J,1}^{(h)} &= (\tilde{\mathbf{Y}}^{(J)}(h+2) - \tilde{\mathbf{Y}}^{(J)}(h+1))(\tilde{\mathbf{X}}(J) - \tilde{\mathbf{Y}}^{(J)}(1))^T, \\ \mathbf{E}_{J,2}^{(h)} &= (\tilde{\mathbf{Y}}^{(J)}(n-2J+1) - \tilde{\mathbf{X}}(n-J))(\tilde{\mathbf{Y}}^{(J)}(n-2J+1-h) - \tilde{\mathbf{Y}}^{(J)}(n-2J-h))^T, \\ \mathbf{p}_{ij}^T \mathbf{E}_{J,1}^{(0)} \mathbf{p}_{ij} &= (\mathbf{p}_{ij}^T (\tilde{\mathbf{Y}}^{(J)}(2) - \tilde{\mathbf{Y}}^{(J)}(1)))^2 - (\mathbf{p}_{ij}^T (\tilde{\mathbf{Y}}^{(J)}(2) - \tilde{\mathbf{X}}(J)))^2, \\ \mathbf{p}_{ij}^T \mathbf{E}_{J,2}^{(0)} \mathbf{p}_{ij} &= (\mathbf{p}_{ij}^T (\tilde{\mathbf{Y}}^{(J)}(n-2J+1) - \tilde{\mathbf{Y}}^{(J)}(n-2J)))^2 - (\mathbf{p}_{ij}^T (\tilde{\mathbf{X}}(n-J) - \tilde{\mathbf{Y}}^{(J)}(n-2J)))^2.\end{aligned}$$

We now present a proposition on the rate of jittering effects from NER-KRVM or NER-KRPVM.

Proposition 1 *Let all the assumptions in Theorem 4 hold. Then with $p/n \rightarrow c \geq 0$,*

$$\begin{aligned}& \frac{\mathbf{p}_{ij}^T \mathbf{E}_J^{(h)} \mathbf{p}_{ij} + 2 \sum_{h=1}^{n-2J} k_H(h) \mathbf{p}_{ij}^T \mathbf{E}_J^{(h)} \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \\ &= O_P(J^{-1} + Jn^{-1} + (J^{-1/2} + J^{1/2}n^{-1/2})(H^{1/2}n^{-1/2} + p_f^{1/2}Hn^{-1} + n^{-1/4})),\end{aligned}$$

where $p_f = 1$ when there are no factors, or $p_f = p$ when there are pervasive factors in the log-price processes.

Proof of Proposition 1. We shall only consider $E_{J,1}^{(h)}$ for $h \geq 0$ in this proof since $E_{J,2}^{(h)}$ can be treated exactly the same with the same rate. Consider

$$\begin{aligned}& \frac{\mathbf{p}_{ij}^T (\tilde{\mathbf{Y}}^{(J)}(1) - \tilde{\mathbf{X}}(J))}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} = I_1 + I_2 + I_3, \text{ where} \\ I_1 &= J^{-1} \sum_{s=0}^{J-1} \frac{\mathbf{p}_{ij}^T (\mathbf{X}(s) - \mathbf{X}(J))}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}}, \quad I_2 = J^{-1} \sum_{s=0}^{J-1} \frac{\mathbf{p}_{ij}^T (e(\mathbf{J}(s)) - e(\mathbf{J}(J)))}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}}, \\ I_3 &= J^{-1} \sum_{s=0}^{J-1} \frac{\mathbf{p}_{ij}^T \epsilon(s)}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}}.\end{aligned}$$

Under Assumption (W1) to (W3) and the rate of jumps removal at $n^{-1/4}$ from Fan and Wang (2007),

$$I_2 = O_P(n^{-1/4}).$$

Also, with Assumption (E3),

$$E_J(I_3)^2 = O(J^{-2} \cdot J + J^{-2} \cdot J^2 \cdot n^{-1}) = O(J^{-1}),$$

so that

$$I_3 = O_P(J^{-1/2}).$$

For I_1 , using the result of Lemma 2 and Assumptions (D1) and (V1),

$$\begin{aligned}I_1 &= J^{-1} \sum_{s=0}^{J-1} \frac{\mathbf{p}_{ij}^T (\mathbf{X}(s) - \mathbf{X}_{v_s})}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} + J^{-1} \sum_{s=0}^{J-1} \frac{\mathbf{p}_{ij}^T (\mathbf{X}_{v_s} - \mathbf{X}_{v_J})}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} + J^{-1} \sum_{s=0}^{J-1} \frac{\mathbf{p}_{ij}^T (\mathbf{X}_{v_J} - \mathbf{X}(J))}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} \\ &= O_P(n^{-1/2}) + O_P(J^{-1} \sum_{s=0}^{J-1} (p_f^{1/2}(J-s)^{1/2}n^{-1} + (J-s)^{1/2}n^{-1/2})) + O_P(n^{-1/2}) \\ &= O_P(p_f^{1/2}J^{1/2}n^{-1} + J^{1/2}n^{-1/2}) = O_P(J^{1/2}n^{-1/2}).\end{aligned}$$

Hence we have

$$\frac{\mathbf{p}_{ij}^T(\tilde{\mathbf{Y}}^{(J)}(1) - \tilde{\mathbf{X}}(J))}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} = O_P(J^{-1/2} + J^{1/2}n^{-1/2} + n^{-1/4}). \quad (\text{S.10})$$

Now consider

$$\begin{aligned} JE &= \mathbf{p}_{ij}^T \mathbf{E}_{J,1}^{(h)} \mathbf{p}_{ij} + 2 \sum_{h=1}^{n-2J} k_H(h) \mathbf{p}_{ij}^T \mathbf{E}_{J,1}^{(h)} \mathbf{p}_{ij} \\ &= [\mathbf{p}_{ij}^T(\tilde{\mathbf{Y}}^{(J)}(1) - \tilde{\mathbf{X}}(J))]^2 + 2\mathbf{p}_{ij}^T(\tilde{\mathbf{Y}}^{(J)}(1) - \tilde{\mathbf{X}}(J)) \cdot R(J), \quad \text{where} \\ R(J) &= \mathbf{p}_{ij}^T(\tilde{\mathbf{X}}(J) - \tilde{\mathbf{Y}}(J+1)) + \sum_{h=1}^{n-2J} k_H(h) \mathbf{p}_{ij}^T(\tilde{\mathbf{Y}}(h+J) - \tilde{\mathbf{Y}}(h+J+1)). \end{aligned}$$

If $k_H(h) = k((h-1)/H)$, then

$$\begin{aligned} R(J) &= \mathbf{p}_{ij}^T \tilde{\mathbf{X}}(J) + \sum_{h=2}^{n-2J} (k_H(h) - k_H(h-1)) \mathbf{p}_{ij}^T \tilde{\mathbf{Y}}(h+J) + k_H(n-2J) \mathbf{p}_{ij}^T \tilde{\mathbf{Y}}(n-J+1) \\ &= \sum_{h=2}^{n-2J} (k_H(h) - k_H(h-1)) \mathbf{p}_{ij}^T(\tilde{\mathbf{Y}}(h+J) - \tilde{\mathbf{X}}(J)) + k_H(n-2J) \mathbf{p}_{ij}^T(\tilde{\mathbf{X}}(J) + \tilde{\mathbf{Y}}(n-J+1)) \\ &= \sum_{h=2}^{n-2J} k'(\xi_{h-1}) H^{-1}(\tilde{\mathbf{Y}}(h+J) - \tilde{\mathbf{X}}(J)) + O_P(n^{-1}), \end{aligned}$$

where $(h-2)/H \leq \xi_{h-1} \leq (h-1)/H$. If $k_H(h) = k(h/H)$, following the above steps, we can show

$$\begin{aligned} R(J) &= \sum_{h=1}^{n-2J} k'(\xi_h) H^{-1}(\tilde{\mathbf{Y}}(h+J) - \tilde{\mathbf{X}}(J)) + O_P(n^{-1}) \\ &= D_1 + D_2 + D_3 + O_P(1), \quad \text{where} \\ D_1 &= \sum_{h=1}^{n-2J} k'(\xi_h) H^{-1} \mathbf{p}_{ij}^T(\mathbf{X}(h+J) - \mathbf{X}(J)), \\ D_2 &= \sum_{h=1}^{n-2J} k'(\xi_h) H^{-1} \mathbf{p}_{ij}^T(e(\mathbf{J}(h+J)) - e(\mathbf{J}(J))), \\ D_3 &= \sum_{h=1}^{n-2J} k'(\xi_h) H^{-1} \mathbf{p}_{ij}^T \boldsymbol{\epsilon}(h+J). \end{aligned}$$

Using Assumption (W1) to (W3) and the rate of jump removal in Fan and Wang (2007),

$$D_2/(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2} = O_P(n^{-1/4}).$$

With Assumption (E3),

$$D_3/(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2} = O_P(H^{-1/2} + n^{-1/2}) = O_P(n^{-1/2}).$$

We further decompose $D_1 = D_{11} + D_{12} + D_{13}$, where

$$\begin{aligned} D_{11} &= \sum_{h=1}^{n-2J} k'(\xi_h) H^{-1} \mathbf{p}_{ij}^T (\mathbf{X}(h+J) - \mathbf{X}_{v_{h+J}}), \\ D_{12} &= \sum_{h=1}^{n-2J} k'(\xi_h) H^{-1} \mathbf{p}_{ij}^T (\mathbf{X}_{v_{h+J}} - \mathbf{X}_{v_J}), \\ D_{13} &= \sum_{h=1}^{n-2J} k'(\xi_h) H^{-1} \mathbf{p}_{ij}^T (\mathbf{X}_{v_J} - \mathbf{X}(J)). \end{aligned}$$

Using (S.4) and Assumption (A4), we have

$$\begin{aligned} \frac{D_{11}}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} &= \sum_{h=1}^{n-2J} k'(\xi_h) H^{-1} (A_d^{ij}(h+J) + A_v^{ij}(h+J)) \\ &= O_P(H^{-1/2} n^{-1/2} + p_f^{1/2} n^{-1}). \end{aligned}$$

Using Lemma 2, we also have

$$\frac{D_{13}}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} = O_P(n^{-1/2}).$$

To find the rate of D_{12} , We decompose $D_{12} = D_{12,d} + D_{12,v}$, where

$$\begin{aligned} \frac{D_{12,d}}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} &= \sum_{h=1}^{n-2J} k'(e_h) H^{-1} \sum_{s=1}^h \frac{\mathbf{p}_{ij}^T \mathbf{A}(v_{s+j-1}, v_{s+J}) \mathbf{Z}_{d,s}^j}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} = O_P(p_f^{1/2} H n^{-1}), \\ \frac{D_{12,v}}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} &= \sum_{h=1}^{n-2J} k'(e_h) H^{-1} \sum_{s=1}^h \frac{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(v_{s+j-1}, v_{s+J})^{1/2} \mathbf{Z}_{v,s}^j}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} \\ &= \sum_{h=1}^{n-2J} k'(e_h) H^{-1} O_P(h^{1/2} n^{-1/2}) = O_P(H^{1/2} n^{-1/2}). \end{aligned}$$

Hence combining the rates obtained,

$$\frac{R(J)}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} = O_P(H^{1/2} n^{-1/2} + p_f^{1/2} H n^{-1} + n^{-1/4}). \quad (\text{S.11})$$

With (S.10) and (S.11), and the fact that the terms involving $\mathbf{E}_{J,2}^{(h)}$ give exactly the same rates as those involving $\mathbf{E}_{J,1}^{(h)}$, the jittering effect has

$$\begin{aligned} &\frac{\mathbf{p}_{ij}^T \mathbf{E}_J^{(h)} \mathbf{p}_{ij} + 2 \sum_{h=1}^{n-2J} k_H(h) \mathbf{p}_{ij}^T \mathbf{E}_J^{(h)} \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \\ &= O_P(J^{-1} + J n^{-1} + n^{-1/4} + (J^{-1/2} + J^{1/2} n^{-1/2})(H^{1/2} n^{-1/2} + p_f^{1/2} H n^{-1} + n^{-1/4})). \quad \square \end{aligned}$$

To finish the proof of Theorem 2, using the notation in the proof of Proposition 1, we decom-

pose

$$\begin{aligned} \frac{\mathbf{p}_{ij}^T \mathbf{K}(\tilde{\mathbf{Y}}^{*(J)}) \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1 &= K_1 + K_{21} + K_{22} + K_3, \text{ with} \\ K_1 &= \frac{\mathbf{p}_{ij}^T \boldsymbol{\gamma}^{(0)}(\tilde{\mathbf{X}}_v^{*(J)}) \mathbf{p}_{ij} + 2 \sum_{h=1}^{n-2J} k_H(h) \mathbf{p}_{ij}^T \boldsymbol{\gamma}^{(h)}(\tilde{\mathbf{X}}_v^{*(J)}) \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1, \\ K_{21} &= \frac{\mathbf{p}_{ij}^T \boldsymbol{\gamma}^{(0)}(\tilde{\mathbf{X}}_v^{*(J)}, \mathbf{E}^{*(J)}) \mathbf{p}_{ij} + 2 \sum_{h=1}^{n-2J} k_H(h) \mathbf{p}_{ij}^T \boldsymbol{\gamma}^{(h)}(\tilde{\mathbf{X}}_v^{*(J)}, \mathbf{E}^{*(J)}) \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ K_{22} &= \frac{\mathbf{p}_{ij}^T \boldsymbol{\gamma}^{(0)}(\mathbf{E}^{*(J)}, \tilde{\mathbf{X}}_v^{*(J)}) \mathbf{p}_{ij} + 2 \sum_{h=1}^{n-2J} k_H(h) \mathbf{p}_{ij}^T \boldsymbol{\gamma}^{(h)}(\mathbf{E}^{*(J)}, \tilde{\mathbf{X}}_v^{*(J)}) \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ K_3 &= \frac{\mathbf{p}_{ij}^T \boldsymbol{\gamma}^{(0)}(\mathbf{E}^{*(J)}) \mathbf{p}_{ij} + 2 \sum_{h=1}^{n-2J} k_H(h) \mathbf{p}_{ij}^T \boldsymbol{\gamma}^{(h)}(\mathbf{E}^{*(J)}) \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \end{aligned}$$

where we define

$$\begin{aligned} \tilde{\mathbf{X}}_{v_s}^{*(J)} &= \mathbf{X}_{v_{s+J-1}} + e(\mathbf{J}(s+J-1)), \\ \mathbf{E}^{*(J)}(s) &= \begin{cases} \mathbf{0}, & s = 1, n-2J+1; \\ (\mathbf{X}(s+J-1) - \mathbf{X}_{v_{s+J-1}}) + \boldsymbol{\epsilon}(s+J-1), & \text{otherwise.} \end{cases}, \end{aligned}$$

and

$$\boldsymbol{\gamma}^{(h)}(\tilde{\mathbf{X}}_v^{*(J)}, \mathbf{E}^{*(J)}) = \sum_{s=h+2}^{n-2J+1} (\tilde{\mathbf{X}}_{v_s}^{*(J)} - \tilde{\mathbf{X}}_{v_{s-1}}^{*(J)}) (\mathbf{E}^{*(J)}(s-h) - \mathbf{E}^{*(J)}(s-h-1))^T,$$

with similar definition for $\boldsymbol{\gamma}^{(h)}(\mathbf{E}^{*(J)}, \tilde{\mathbf{X}}_v^{*(J)})$.

Proposition 2 *Let all the assumptions in Theorem 4 hold. Then with $p/n \rightarrow c \geq 0$,*

$$K_1 = O_P(p_f^{1/2} H^{1/2} n^{-1/2} + p_f^{1/2} n^{-1/4}),$$

where $p_f = 1$ when there are no factors, or $p_f = p$ when there are pervasive factors in the log-price processes.

Proof of Proposition 2. Firstly, by Lemma 1, since $J = o(n)$, we immediately have

$$\left| \frac{\mathbf{p}_{ij}^T \boldsymbol{\gamma}^{(0)}(\tilde{\mathbf{X}}_v^{*(J)}) \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1 \right| = O_P(1 \cdot n^{-1/2} + p_f^{1/2} n^{-1/2}) = O_P(p_f^{1/2} n^{-1/2}).$$

Also, defining

$$\begin{aligned} z_{d,s}^{ij} &= \frac{\mathbf{p}_{ij}^T \mathbf{A}(v_{s+J-2}, v_{s+J-1}) \mathbf{Z}_{d,s+J-1}^j}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}}, \quad z_{v,s}^{ij} = \frac{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(v_{s+J-2}, v_{s+J-1})^{1/2} \mathbf{Z}_{v,s+J-1}^j}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}}, \\ e_s^{ij} &= \frac{\mathbf{p}_{ij}^T (e(\mathbf{J}(s+J-1)) - e(\mathbf{J}(s+J-2)))}{(\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}}, \end{aligned}$$

we can decompose (we omit any superscripts ij if no ambiguity arises)

$$\begin{aligned} \frac{\sum_{h=1}^{n-2J} k_H(h) \mathbf{p}_{ij}^T \gamma^{(h)}(\tilde{\mathbf{X}}_v^{*(J)}) \mathbf{p}_{ij}}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} &= \sum_{r=1}^7 M_r, \text{ where} \\ M_1 &= \sum_{s=3}^{n-2J+1} z_{d,s} \sum_{h=1}^{s-2} k_H(h) z_{d,s-h}, \quad M_2 = \sum_{s=3}^{n-2J+1} z_{d,s} \sum_{h=1}^{s-2} k_H(h) z_{v,s-h}, \\ M_3 &= \sum_{s=3}^{n-2J+1} z_{v,s} \sum_{h=1}^{s-2} k_H(h) z_{d,s-h}, \quad M_4 = \sum_{s=3}^{n-2J+1} z_{v,s} \sum_{h=1}^{s-2} k_H(h) z_{v,s-h}, \\ M_5 &= \sum_{s=3}^{n-2J+1} e_s \sum_{h=1}^{s-2} k_H(h) (z_{d,s-h} + z_{v,s-h}), \\ M_6 &= \sum_{s=3}^{n-2J+1} (z_{d,s} + z_{v,s}) \sum_{h=1}^{s-2} k_H(h) e_{s-h}, \quad M_7 = \sum_{s=3}^{n-2J+1} e_s \sum_{h=1}^{s-2} k_H(h) e_{s-h}. \end{aligned}$$

Using Assumption (D1), we have

$$M_1 = O_P(p_f H n^{-1}).$$

Using Assumption (D1) and (V1),

$$M_2 = O_P(p_f^{1/2} H^{1/2} n^{-1/2}).$$

Using the Burkholder's inequality,

$$M_3 = O_P(n \cdot n^{-1} \cdot p_f H^2 n^{-2})^{1/2} = O_P(p_f^{1/2} H n^{-1}).$$

Using the Burkholder's inequality again,

$$M_4 = O_P(n \cdot n^{-1} \cdot H n^{-1})^{1/2} = O_P(H^{1/2} n^{-1/2}).$$

Finally, with Assumption (W1) to (W3), using the jump removal rate $n^{-1/4}$ from Fan and Wang (2007),

$$M_5 = O_P(n^{-1/4} \cdot (p_f^{1/2} H n^{-1} + H^{1/2} n^{-1/2})), \quad M_6 = O_P(p_f^{1/2} n^{-1/4} + n^{-1/4}), \quad M_7 = O_P(n^{-1/2}).$$

Combining all the rates calculated, we can conclude that

$$K_1 = O_P(p_f^{1/2} H^{1/2} n^{-1/2} + p_f^{1/2} n^{-1/4}). \quad \square$$

Proposition 3 *Let all the assumptions in Theorem 4 hold. Then with $p/n \rightarrow c \geq 0$,*

$$K_3 = \begin{cases} O_P(H^{-3/2} n^{1/2}), & k_H(h) = k((h-1)/H); \\ O_P(H^{-2} n), & k_H(h) = k(h/H). \end{cases}$$

Proof of Proposition 3. Firstly, define

$$r_s^{ij} = \frac{\mathbf{p}_{ij}^T \mathbf{E}^{*(J)}(s)}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}},$$

so that by the definition of $\mathbf{E}^{*(J)}(s)$, $r_1^{ij} = r_{n-2J+1}^{ij} = 0$.

If no ambiguity arises, we drop the superscript in r_s^{ij} for easier presentation. Consider the decomposition

$$\begin{aligned}
& \frac{\mathbf{p}_{ij}^T \gamma^{(0)}(\mathbf{E}^{*(J)}) \mathbf{p}_{ij} + 2 \sum_{h=1}^{n-2J+1} k_H(h) \mathbf{p}_{ij}^T \gamma^{(h)}(\mathbf{E}^{*(J)}) \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \\
&= \sum_{s=2}^{n-2J+1} (r_s - r_{s-1})^2 + 2 \sum_{h=1}^{n-2J} k_H(h) \sum_{s=h+2}^{n-2J+1} (r_s - r_{s-1})(r_{s-h} - r_{s-h-1}) \\
&= \sum_{s=2}^{n-2J+1} (r_s - r_{s-1})^2 + 2 \sum_{s=2}^{n-2J+1} (r_s - r_{s-1}) \sum_{h=1}^{s-2} k_H(h)(r_{s-h} - r_{s-h-1}) \\
&= \sum_{s=2}^{n-2J+1} (r_s - r_{s-1}) \left(r_s - r_{s-1} + 2 \sum_{h=1}^{s-2} k_H(h)(r_{s-h} - r_{s-h-1}) \right) \\
&= r_2^2 + \sum_{s=3}^{n-2J+1} (r_s - r_{s-1}) \left((r_s + r_{s-1}) + (2k_H(1)r_{s-1} - 2r_{s-1}) + 2 \sum_{h=2}^{s-2} (k_H(h) - k_H(h-1))r_{s-h} \right) \\
&= IE_1 + IE_2 + IE_3, \text{ where} \\
IE_1 &= \sum_{s=2}^{n-2J+1} (r_s^2 - r_{s-1}^2), \quad IE_2 = 2(k_H(1) - 1) \sum_{s=3}^{n-2J+1} (r_s - r_{s-1})r_{s-1}, \\
IE_3 &= 2 \sum_{s=4}^{n-2J+1} (r_s - r_{s-1}) \sum_{h=2}^{s-2} k'(\xi_h) H^{-1} r_{s-h},
\end{aligned}$$

where $(h-2)/H \leq \xi_h \leq (h-1)/H$ if $k_H(h) = k((h-1)/H)$, and $(h-1)/H \leq \xi_h \leq h/H$ if $k_H(h) = k(h/H)$.

Firstly, since $r_1 = r_{n-2J+1} = 0$, we have

$$IE_1 = 0.$$

Secondly, $IE_2 = 0$ if $k_H(h) = k((h-1)/H)$ since then $k_H(1) = k(0) = 1$.

For $k_H(h) = k(h/H)$, using the fact that $k'(0) = 0$,

$$IE_2 = 2k'(\xi_1)H^{-1} \sum_{s=3}^{n-2J+1} (r_s - r_{s-1})r_{s-1} = 2k''(\xi'_1)\xi_1 H^{-1} \sum_{s=3}^{n-2J+1} (r_s - r_{s-1})r_{s-1}, \quad 0 \leq \xi'_1 \leq \xi_1 \leq H^{-1}.$$

Decompose further $r_s = r_{x,s} + r_{\epsilon,s}$, where

$$r_{x,s} = \frac{\mathbf{p}_{ij}^T (\mathbf{X}(s) - \mathbf{X}_{v_s})}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}}, \quad r_{\epsilon,s} = \frac{\mathbf{p}_{ij}^T \epsilon(s)}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}}.$$

Then by Lemma 2 and Assumption (E1), we have

$$IE_2 \leq 8H^{-2}k''(\xi'_1) \sum_{s=3}^{n-2J+1} (r_{x,s-1}^2 + r_{\epsilon,s-1}^2) = O_P(H^{-2}(1+n)) = O_P(H^{-2}n).$$

Hence we have

$$IE_2 = \begin{cases} 0, & k_H(h) = k((h-1)/H); \\ O_P(H^{-2}n), & k_H(h) = k(h/H). \end{cases}$$

We decompose

$$\begin{aligned}
IE_3 &= 2 \sum_{s=4}^{n-2J+1} r_s \sum_{h=2}^{s-2} k'(\xi_h) H^{-1} r_{s-h} - 2 \sum_{s=3}^{n-2J} r_s \sum_{h=2}^{s-1} k'(\xi_h) H^{-1} r_{s-h+1} \\
&= 2 \sum_{s=3}^{n-2J} r_s \sum_{h=2}^{s-1} k'(\xi_h) H^{-1} (r_{s-h} - r_{s-h+1}) \\
&= 2 \sum_{s=3}^{n-2J} r_s \left(\sum_{h=2}^{s-1} k'(\xi_h) H^{-1} r_{s-h} - \sum_{h=1}^{s-2} k'(\xi_{h+1}) H^{-1} r_{s-h} \right) \\
&= 2 \sum_{s=3}^{n-2J} r_s \left(\sum_{h=1}^{s-2} k''(\xi'_h) (\xi_h - \xi_{h+1}) H^{-1} r_{s-h} - k'(\xi_1) H^{-1} r_{s-1} \right), \quad \xi_h \leq \xi'_h \leq \xi_{h+1} \\
&= IE_{3,1} + IE_{3,2}, \text{ where} \\
IE_{3,1} &= -2 \sum_{s=3}^{n-2J} k'(\xi_1) H^{-1} r_s r_{s-1}, \quad IE_{3,2} = 2 \sum_{s=3}^{n-2J} r_s \sum_{h=1}^{s-2} k''(\xi'_h) (\xi_h - \xi_{h+1}) H^{-1} r_{s-h}.
\end{aligned}$$

For $k_H(h) = k((h-1)/H)$, $\xi_1 = 0$ with $k'(\xi_1) = k_H(1) - 1 = 0$, and hence $IE_{3,1} = 0$. For $k_H(h) = k(h/H)$, using Lemma 2 and Assumption (E3),

$$\begin{aligned}
IE_{3,1} &= -2 \sum_{s=3}^{n-2J} k''(\xi'_1) \xi_1 H^{-1} r_s r_{s-1} \\
&= O_P(H^{-1}n^{-1} + (H^{-3}n^{-1} + H^{-2}n^{-1}n^{-1})^{1/2} + (H^{-3} + H^{-2}n^{-1})^{1/2}) = O_P(H^{-3/2}).
\end{aligned}$$

For $IE_{3,2}$, we have

$$\begin{aligned}
IE_{3,2} &= O_P(p_f H^{-1}n^{-1} + (n \cdot n^{-1} H^{-3}n^{-1})^{1/2} + (nH^{-3})^{1/2}) = O_P(p_f H^{-1}n^{-1} + H^{-3/2}n^{1/2}) \\
&= O_P(H^{-3/2}n^{1/2}),
\end{aligned}$$

which gives $IE_3 = O_P(H^{-3/2}n^{1/2})$. Hence combining the rates for IE_1 to IE_3 , we have

$$K_3 = \begin{cases} O_P(0 + 0 + H^{-3/2}n^{1/2}) = O_P(H^{-3/2}n^{1/2}), & k_H(h) = k((h-1)/H); \\ O_P(0 + H^{-2}n + H^{-3/2}n^{1/2}) = O_P(H^{-2}n), & k_H(h) = k(h/H). \end{cases} \quad \square$$

Proposition 4 *Let all the assumptions in Theorem 4 hold. Then with $p/n \rightarrow c \geq 0$,*

$$K_{21} + K_{22} = O_P(H^{-1/2} + n^{-1/4} + p_f^{1/2}n^{-1/2} + H^{-1}n^{1/4} + J^{1/2}n^{-1/2}),$$

where $p_f = 1$ when there are no factors, or $p_f = p$ when there are pervasive factors in the log-price processes.

Proof of Proposition 4. Using the notations used in the proof of Proposition 3, dropping superscripts ij in r_s^{ij} , we can decompose

$$\begin{aligned}
K_{21} &= \sum_{s=2}^{n-2J+1} \mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_s}^{*(J)} - \tilde{\mathbf{X}}_{v_{s-1}}^{*(J)}) \left(r_s - r_{s-1} + 2 \sum_{h=1}^{s-2} k_H(h) (r_{s-h} - r_{s-h-1}) \right) \\
&= \sum_{s=2}^{n-2J+1} \mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_s}^{*(J)} - \tilde{\mathbf{X}}_{v_{s-1}}^{*(J)}) \left((r_s + r_{s-1}) + 2(k_H(1) - 1)r_{s-1} + 2 \sum_{h=2}^{s-2} (k_H(h) - k_H(h-1))r_{s-h} \right).
\end{aligned}$$

Similarly,

$$\begin{aligned}
K_{22} &= \sum_{s=2}^{n-2J+1} (r_s - r_{s-1}) \left(\mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_s}^{*(J)} - \tilde{\mathbf{X}}_{v_{s-1}}^{*(J)}) + 2 \sum_{h=1}^{s-2} k_H(h) \mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_{s-h}}^{*(J)} - \tilde{\mathbf{X}}_{v_{s-h-1}}^{*(J)}) \right) \\
&= \sum_{s=2}^{n-2J+1} (r_s - r_{s-1}) \left(\mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_s}^{*(J)} + \tilde{\mathbf{X}}_{v_{s-1}}^{*(J)}) + 2(k_H(1) - 1) \mathbf{p}_{ij}^T \tilde{\mathbf{X}}_{v_{s-1}} \right. \\
&\quad \left. + \left\{ 2 \sum_{h=2}^{s-2} (k_H(h) - k_H(h-1)) \mathbf{p}_{ij}^T \tilde{\mathbf{X}}_{v_{s-h}}^{*(J)} - 2k_H(s-2) \mathbf{p}_{ij}^T \tilde{\mathbf{X}}_{v_1}^{*(J)} \right\} \right).
\end{aligned}$$

Hence we have $K_{21} + K_{22} = \sum_{i=1}^6 CE_i$, where

$$\begin{aligned}
CE_1 &= 2 \sum_{s=2}^{n-2J+1} \mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_s}^{*(J)} r_s - \tilde{\mathbf{X}}_{v_{s-1}}^{*(J)} r_{s-1}), \\
CE_2 &= 2(k_H(1) - 1) \sum_{s=3}^{n-2J+1} \mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_s}^{*(J)} - \tilde{\mathbf{X}}_{v_{s-1}}^{*(J)}) r_{s-1}, \\
CE_3 &= 2(k_H(1) - 1) \sum_{s=3}^{n-2J+1} \mathbf{p}_{ij}^T \tilde{\mathbf{X}}_{v_{s-1}}^{*(J)} (r_s - r_{s-1}), \\
CE_4 &= 2 \sum_{s=4}^{n-2J+1} \mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_s}^{*(J)} - \tilde{\mathbf{X}}_{v_{s-1}}^{*(J)}) \sum_{h=2}^{s-2} k'(\xi_h) H^{-1} r_{s-h}, \\
CE_5 &= 2 \sum_{s=4}^{n-2J+1} (r_s - r_{s-1}) \sum_{h=2}^{s-2} k'(\xi_h) H^{-1} \mathbf{p}_{ij}^T \tilde{\mathbf{X}}_{v_{s-h}}^{*(J)}, \\
CE_6 &= -2 \sum_{s=3}^{n-2J+1} (r_s - r_{s-1}) k_H(s-2) \mathbf{p}_{ij}^T \tilde{\mathbf{X}}_{v_1}^{*(J)}.
\end{aligned}$$

Using $r_1 = r_{n-2J+1} = 0$, we have

$$CE_1 = 0.$$

For $k_H(h) = k((h-1)/H)$, $CE_2 = CE_3 = 0$. Otherwise, using the notations in the proof of Proposition 2 and dropping the superscripts ij in $z_{d,s}^{ij}$, $z_{v,s}^{ij}$ and e_s^{ij} when no ambiguity arises,

$$\begin{aligned}
CE_2 &= 2k''(\xi'_1) \xi_1 H^{-1} \sum_{s=3}^{n-2J+1} (z_{d,s} + z_{v,s} + e_s) r_{s-1}, \quad 0 \leq \xi'_1 \leq \xi_1 \leq H^{-1} \\
&= O_P(H^{-2} \cdot (p_f^{1/2} n^{-1/2} + n^{1/2} \cdot n^{-1/2} + n^{-1/4})) = O_P(H^{-2}).
\end{aligned}$$

where we used Assumption (D1), (V1), (E3) and Lemma 2. Similarly,

$$\begin{aligned}
CE_3 &= 2k''(\xi'_1) \xi_1 H^{-1} \left(\sum_{s=3}^{n-2J+1} \mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_{s-1}}^{*(J)} - \tilde{\mathbf{X}}_{v_s}^{*(J)}) r_s - \mathbf{p}_{ij}^T \tilde{\mathbf{X}}_{v_2}^{*(J)} r_2 \right) \\
&= O_P(H^{-2} + H^{-2}(J^{1/2} n^{-1/2} + n^{-1/4})) = O_P(H^{-2}).
\end{aligned}$$

We also have

$$CE_4 = 2 \sum_{s=4}^{n-2J+1} (z_{d,s} + z_{v,s} + e_s) \sum_{h=2}^{s-2} k'(\xi_h) H^{-1} r_{s-h} = CE_{4,d} + CE_{4,v} + CE_{4,e},$$

where

$$\begin{aligned} CE_{4,d} &= O_P(p_f n^{-1} + p_f^{1/2} H^{-1/2} n^{-1/2} + p_f^{1/2} H^{-1/2}) = O_P(p_f^{1/2} H^{-1/2}), \\ CE_{4,v} &= O_P(n \cdot n^{-1} \cdot (H^{-3} + H^{-2} n^{-1} + H^{-2} + H^{-1} n^{-1} + n^{-1})^{1/2})^{1/2} = O_P(H^{-1/2} + n^{-1/4}), \\ CE_{4,e} &= O_P(H^{-1/2} n^{-1/4}). \end{aligned}$$

Hence

$$CE_4 = O_P(H^{-1/2} + n^{-1/4}).$$

For CE_5 , we have

$$\begin{aligned} CE_5 &= 2 \sum_{s=4}^{n-2J+1} r_s \sum_{h=2}^{s-2} k'(\xi_h) H^{-1} \mathbf{p}_{ij}^T \tilde{\mathbf{X}}_{v_{s-h}}^{*(J)} - 2 \sum_{s=3}^{n-2J} r_s \sum_{h=2}^{s-1} k'(\xi_h) H^{-1} \mathbf{p}_{ij}^T \tilde{\mathbf{X}}_{v_{s-h+1}}^{*(J)} \\ &= -2r_3 k'(\xi_2) H^{-1} \mathbf{p}_{ij}^T \tilde{\mathbf{X}}_{v_2}^{*(J)} \\ &\quad + 2 \sum_{s=4}^{n-2J} r_s \left(\sum_{h=2}^{s-2} k'(\xi_h) H^{-1} \mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_{s-h}}^{*(J)} - \tilde{\mathbf{X}}_{v_{s-h+1}}^{*(J)}) - k'(\xi_{s-1}) H^{-1} \mathbf{p}_{ij}^T \tilde{\mathbf{X}}_{v_2}^{*(J)} \right) \\ &= O_P(H^{-1} (J^{1/2} n^{-1/2} + n^{-1/4}) + (n \cdot p_f n^{-2} + n \cdot H^{-1} n^{-1} + n \cdot H^{-2} n^{-1/2})^{1/2} \\ &\quad + H^{-1/2} (J^{1/2} n^{-1/2} + n^{-1/4})) \\ &= O_P(p_f^{1/2} n^{-1/2} + H^{-1/2} + H^{-1} n^{1/4}). \end{aligned}$$

Finally,

$$\begin{aligned} CE_6 &= \left(-2 \sum_{s=3}^{n-2J} r_s k_H(s-2) + 2 \sum_{s=2}^{n-2J} r_s k_H(s-1) \right) \mathbf{p}_{ij}^T \tilde{\mathbf{X}}_{v_1}^{*(J)} \\ &= 2 \mathbf{p}_{ij}^T \tilde{\mathbf{X}}_{v_1}^{*(J)} \left(\sum_{s=3}^{n-2J} r_s k'(\xi_{s-1}) H^{-1} + r_2 k_H(1) \right) \\ &= O_P((J^{1/2} n^{-1/2} + n^{-1/4})(H^{-1/2} + 1)) = O_P(J^{1/2} n^{-1/2} + n^{-1/4}). \end{aligned}$$

Hence

$$K_{21} + K_{22} = O_P(H^{-1/2} + n^{-1/4} + p_f^{1/2} n^{-1/2} + H^{-1} n^{1/4} + J^{1/2} n^{-1/2}). \quad \square$$

To finish the proof of Theorem 2, we can combine the rates from Proposition 1 to 4 to arrive at

$$\begin{aligned} &\max_{\substack{i=1,\dots,p \\ j=1,\dots,L}} \left| \frac{\mathbf{p}_{ij}^T \tilde{\Sigma}(\tau_{j-1}, \tau_j)^K \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1 \right| \\ &= O_P(J^{-1} + J^{1/2} n^{-1/2} + p_f^{1/2} n^{-1/4} + p_f^{1/2} H^{1/2} n^{-1/2} + H^{-3/2} n^{1/2} + H^{-1/2} + H^{-1} n^{1/4}). \end{aligned}$$

The optimized rate is achieved using $H \asymp n^{1/2}$, with the smallest order of J being $J \asymp p_f^{-1/2} n^{1/4}$, and rate of convergence at $p_f^{1/2} n^{-1/4}$. Hence when there are pervasive factors, we need $p = o(n^{1/2})$ for guaranteed convergence. Also, we have

$$\begin{aligned} &\max_{\substack{i=1,\dots,p \\ j=1,\dots,L}} \left| \frac{\mathbf{p}_{ij}^T \tilde{\Sigma}(\tau_{j-1}, \tau_j)^{KP} \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1 \right| \\ &= O_P(J^{-1} + J^{1/2} n^{-1/2} + p_f^{1/2} n^{-1/4} + p_f^{1/2} H^{1/2} n^{-1/2} + H^{-2} n + H^{-1/2} + H^{-1} n^{1/4}). \end{aligned}$$

The optimized rate is achieved using $H \asymp p_f^{-1/5} n^{3/5}$, with the smallest order of J being $J \asymp p_f^{-2/5} n^{1/5}$, and rate of convergence at $p_f^{2/5} n^{-1/5}$. Hence when there are pervasive factors, we need $p = o(n^{1/2})$ for guaranteed convergence.

For $\hat{\Sigma}(0, 1)$, we use the same argument at the end of the proof of Theorem 1 to arrive at

$$\|\hat{\Sigma}(0, 1)^K \Sigma_{\text{Ideal}}(0, 1) - \mathbf{I}_p\| = O_P(p_f^{1/2} n^{-1/4}),$$

where $H \asymp n^{1/2}$ and $J \asymp p_f^{-1/2} n^{1/4}$. Same treatment apply for $\hat{\Sigma}(0, 1)^{KP}$. This completes the proof of the theorem. $\square$

Proof of the Theorem 3

We first present the proof for $\tilde{\Sigma}(\tau_{j-1}, \tau_j)^{PP} = \mathbf{P}(\tilde{\mathbf{Y}})_j$. Let

$$\mathbf{P}(\tilde{\mathbf{Y}})_j = \frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} \tilde{\mathbf{Y}}_s \tilde{\mathbf{Y}}_s^T, \text{ where } \tilde{\mathbf{Y}}_s = \sum_{l=1}^{Q-1} g\left(\frac{l}{Q}\right) (\tilde{\mathbf{Y}}_{s+l} - \tilde{\mathbf{Y}}_{s+l-1}).$$

Then we have

$$\begin{aligned} \mathbf{P}(\tilde{\mathbf{Y}})_j &= \mathbf{P}(\tilde{\mathbf{X}})_j + \mathbf{P}(\tilde{\mathbf{X}}, \mathbf{E})_j + \mathbf{P}(\mathbf{E}, \tilde{\mathbf{X}})_j + \mathbf{P}(\mathbf{E})_j, \text{ where} \\ \mathbf{P}(\tilde{\mathbf{X}})_j &= \frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} \tilde{\mathbf{X}}_{v_s} \tilde{\mathbf{X}}_{v_s}^T, \text{ with } \tilde{\mathbf{X}}_s = \sum_{l=1}^{Q-1} g\left(\frac{l}{Q}\right) (\tilde{\mathbf{X}}_{v_{s+l}} - \tilde{\mathbf{X}}_{v_{s+l-1}}), \\ \mathbf{P}(\mathbf{E})_j &= \frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} \bar{\mathbf{E}}(s) \bar{\mathbf{E}}(s)^T, \text{ with } \bar{\mathbf{E}}_s = \sum_{l=1}^{Q-1} g\left(\frac{l}{Q}\right) (\mathbf{E}(s+l) - \mathbf{E}(s+l-1)), \\ \mathbf{P}(\tilde{\mathbf{X}}, \mathbf{E})_j &= \frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} \tilde{\mathbf{X}}_{v_s} \bar{\mathbf{E}}(s)^T, \mathbf{P}(\mathbf{E}, \tilde{\mathbf{X}}) = \mathbf{P}(\tilde{\mathbf{X}}, \mathbf{E})^T. \end{aligned}$$

Hence

$$\begin{aligned} \mathbf{p}_{ij}^T \mathbf{P}(\tilde{\mathbf{Y}})_j \mathbf{p}_{ij} &= \mathbf{p}_{ij}^T \mathbf{P}(\tilde{\mathbf{X}})_j \mathbf{p}_{ij} + 2\mathbf{p}_{ij}^T \mathbf{P}(\tilde{\mathbf{X}}, \mathbf{E})_j \mathbf{p}_{ij} + \mathbf{p}_{ij}^T \mathbf{P}(\mathbf{E})_j \mathbf{p}_{ij} \\ &= I_1^{PP} + 2I_2^{PP} + I_3^{PP}. \end{aligned} \tag{S.12}$$

By the definition of ψ , as $Q \rightarrow \infty$, standard rate of convergence of Riemann integral implies that

$$\left| \frac{1}{\psi Q} \sum_{l=1}^{Q-1} g\left(\frac{l}{Q}\right)^2 - 1 \right| = O(Q^{-1}). \tag{S.13}$$

We further decompose

$$\begin{aligned} I_1^{PP} &= \mathbf{p}_{ij}^T \mathbf{P}(\tilde{\mathbf{X}})_j \mathbf{p}_{ij} \\ &= \mathbf{p}_{ij}^T \left(\frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} \left(\sum_{l=0}^{Q-1} g\left(\frac{l}{Q}\right) (\tilde{\mathbf{X}}_{v_{s+l}} - \tilde{\mathbf{X}}_{v_{s+l-1}}) \sum_{l'=0}^{Q-1} g\left(\frac{l'}{Q}\right) (\tilde{\mathbf{X}}_{v_{s+l'}} - \tilde{\mathbf{X}}_{v_{s+l'-1}})^T \right) \right) \mathbf{p}_{ij} \\ &= \frac{1}{\psi Q} \sum_{l=1}^{Q-1} g^2\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^T \sum_{s=1}^{n-Q+1} (\tilde{\mathbf{X}}_{v_{s+l}} - \tilde{\mathbf{X}}_{v_{s+l-1}}) (\tilde{\mathbf{X}}_{v_{s+l}} - \tilde{\mathbf{X}}_{v_{s+l-1}})^T \mathbf{p}_{ij} \\ &\quad + \frac{1}{\psi Q} \sum_{l \neq l'} g\left(\frac{l}{Q}\right) g\left(\frac{l'}{Q}\right) \mathbf{p}_{ij}^T \sum_{s=1}^{n-Q+1} (\tilde{\mathbf{X}}_{v_{s+l}} - \tilde{\mathbf{X}}_{v_{s+l-1}}) (\tilde{\mathbf{X}}_{v_{s+l'}} - \tilde{\mathbf{X}}_{v_{s+l'-1}})^T \mathbf{p}_{ij}, \\ &= I_{1,1}^{PP} + I_{1,2}^{PP}. \end{aligned}$$

Then by Lemma 1 and (S.13),

$$\left| \frac{I_{1,1}^{PP}}{\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1 \right| = O_P(p_f^{1/2} n^{-1/2} + Q^{-1}). \quad (\text{S.14})$$

For $I_{1,2}^{PP}$,

$$\begin{aligned} I_{1,2}^{PP} &= 2(\psi Q)^{-1} \sum_{s=1}^{n-Q+1} \sum_{l=2}^{Q-1} g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^\top (\tilde{\mathbf{X}}_{v_{s+l}} - \tilde{\mathbf{X}}_{v_{s+l-1}}) \sum_{l'=1}^{l-1} g\left(\frac{l'}{Q}\right) \mathbf{p}_{ij}^\top (\tilde{\mathbf{X}}_{v_{s+l'}} - \tilde{\mathbf{X}}_{v_{s+l'-1}}) \\ &= 2(\psi Q)^{-1} \sum_{s=1}^{n-Q+1} \sum_{l=2}^{Q-1} g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^\top (\mathbf{X}_{v_{s+l}} - \mathbf{X}_{v_{s+l-1}} + e(\mathbf{J}_{v_{s+l}}) - e(\mathbf{J}_{v_{s+l-1}})) \\ &\quad \cdot \sum_{l'=1}^{l-1} g\left(\frac{l'}{Q}\right) \mathbf{p}_{ij}^\top (\mathbf{X}_{v_{s+l'}} - \mathbf{X}_{v_{s+l'-1}} + e(\mathbf{J}_{v_{s+l'}}) - e(\mathbf{J}_{v_{s+l'-1}})) \\ &= 2(\psi Q)^{-1} (P_1 + P_2 + P_3 + P_4), \end{aligned}$$

where

$$\begin{aligned} P_1 &= \sum_{s=1}^{n-Q+1} \sum_{l=2}^{Q-1} g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^\top (\mathbf{X}_{v_{s+l}} - \mathbf{X}_{v_{s+l-1}}) \sum_{l'=1}^{l-1} g\left(\frac{l'}{Q}\right) \mathbf{p}_{ij}^\top (\mathbf{X}_{v_{s+l'}} - \mathbf{X}_{v_{s+l'-1}}), \\ P_2 &= \sum_{s=1}^{n-Q+1} \sum_{l=2}^{Q-1} g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^\top (\mathbf{X}_{v_{s+l}} - \mathbf{X}_{v_{s+l-1}}) \sum_{l'=1}^{l-1} g\left(\frac{l'}{Q}\right) \mathbf{p}_{ij}^\top (e(\mathbf{J}_{v_{s+l'}}) - e(\mathbf{J}_{v_{s+l'-1}})), \\ P_3 &= \sum_{s=1}^{n-Q+1} \sum_{l=2}^{Q-1} g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^\top (e(\mathbf{J}_{v_{s+l}}) - e(\mathbf{J}_{v_{s+l-1}})) \sum_{l'=1}^{l-1} g\left(\frac{l'}{Q}\right) \mathbf{p}_{ij}^\top (\mathbf{X}_{v_{s+l'}} - \mathbf{X}_{v_{s+l'-1}}), \\ P_4 &= \sum_{s=1}^{n-Q+1} \sum_{l=2}^{Q-1} g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^\top (e(\mathbf{J}_{v_{s+l}}) - e(\mathbf{J}_{v_{s+l-1}})) \sum_{l'=1}^{l-1} g\left(\frac{l'}{Q}\right) \mathbf{p}_{ij}^\top (e(\mathbf{J}_{v_{s+l'}}) - e(\mathbf{J}_{v_{s+l'-1}})). \end{aligned}$$

We focus on P_1 first. Defining

$$A_s(l) = g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^\top \mathbf{A}(v_{s+l-1}, v_{s+l}) \mathbf{Z}_{d,s+l}^j, \quad V_s(l) = g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(v_{s+l-1}, v_{s+l})^{1/2} \mathbf{Z}_{v,s+l}^j,$$

we can decompose $P_1 = P_{a,a} + P_{a,v} + P_{v,a} + P_{v,v}$, where

$$P_{a,v} = \sum_{s=1}^{n-Q+1} \sum_{l=2}^{Q-1} A_s(l) \sum_{l'=1}^{l-1} V_s(l'), \quad P_{v,a} = \sum_{s=1}^{n-Q+1} \sum_{l=2}^{Q-1} V_s(l) \sum_{l'=1}^{l-1} A_s(l'),$$

and the other two terms are defined similarly. Since $E_j(A_s(l)) = 0$, and

$$\begin{aligned} E_j \left(\sum_{l=2}^{Q-1} A_s(l) \right)^2 &= \sum_{l=2}^{Q-1} g^2 \left(\frac{l}{Q} \right) \mathbf{p}_{ij}^\top \mathbf{A}(v_{s+l-1}, v_{s+l}) \mathbf{A}(v_{s+l-1}, v_{s+l})^\top \mathbf{p}_{ij} \\ &\quad + \sum_{l \neq l'} g \left(\frac{l}{Q} \right) g \left(\frac{l'}{Q} \right) E_j (\mathbf{p}_{ij}^\top \mathbf{A}(v_{s+l-1}, v_{s+l}) \mathbf{Z}_{d,s+l}^j \mathbf{Z}_{d,s+l'}^j \mathbf{A}(v_{s+l'-1}, v_{s+l'})^\top \mathbf{p}_{ij}) \\ &= O(Q p_f n^{-2} + Q^2 p_f n^{-2}) = O(Q^2 p_f n^{-2}), \end{aligned}$$

we have

$$P_{a,a} = O_P(n \cdot (p_f^{1/2} Q n^{-1})^2) = O_P(p_f Q^2 n^{-1}).$$

Also, Burkholder's inequality can be applied so that for a generic constant C which can change values from line to line,

$$E_j \left(\sum_{l=2}^{Q-1} V_s(l) \right)^2 \leq C \sum_{l=2}^{Q-1} g^2 \left(\frac{l}{Q} \right) \mathbf{p}_{ij}^T \boldsymbol{\Sigma}(v_{s+l-1}, v_{s+l}) \mathbf{p}_{ij} = O(Q p_f n^{-1}),$$

so that

$$P_{a,v} = O_P(n \cdot p_f^{1/2} Q n^{-1} \cdot p_f^{1/2} Q^{1/2} n^{-1/2}) = O_P(p_f Q^{3/2} n^{-1/2}).$$

Since $E(V_s(l) \sum_{l'=1}^{l-1} A_s(l') | \mathcal{F}_{-j} \cup \mathcal{F}_{v_{s+l-1}}^j) = 0$, we can use Burkholder's inequality so that

$$\begin{aligned} E_j \left(\sum_{s=1}^{n-Q+1} V_s(l) \sum_{l'=1}^{l-1} A_s(l') \right)^2 &\leq C \sum_{s=1}^{n-Q+1} E_j^{1/2}(V_s(l)^4) E_j^{1/2} \left(\sum_{l'=1}^{l-1} A_s(l') \right)^4 \\ &= O(n \cdot p_f n^{-1} \cdot Q^2 p_f n^{-2}) = O(p_f^2 Q^2 n^{-2}). \end{aligned}$$

where the second inequality used Lemma 2.7 of Bai and Silverstein (1998). It means that

$$P_{v,a} = O_P(Q \cdot p_f Q n^{-1}) = O_P(p_f Q^2 n^{-1}).$$

Also, similar to the arguments above,

$$E_j \left(\sum_{s=1}^{n-Q+1} V_s(l) \sum_{l'=1}^{l-1} V_s(l') \right)^2 = O(n \cdot p_f n^{-1} \cdot Q p_f n^{-1}) = O(p_f^2 Q n^{-1}).$$

Hence

$$P_{v,v} = O_P(p_f Q^{3/2} n^{-1/2}).$$

With these results, we have

$$P_1 / \mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij} = O_P(p_f Q^2 n^{-1} + p_f^{1/2} Q^{3/2} n^{-1/2}).$$

Using Assumption (W1) to (W3) and the rate of jump removal at $n^{-1/4}$ from Fan and Wang (2007), we have

$$\begin{aligned} \frac{P_2}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \frac{P_3}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} &= O_P(p_f^{1/2} Q^2 n^{-5/4} + Q^{3/2} n^{-3/4}), \\ \frac{P_4}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} &= O_P(Q n^{-1/2}). \end{aligned}$$

Hence using the rates for P_1 to P_4 found above, together with (S.14), we have

$$\left| \frac{I_1^{PP}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1 \right| = O_P(p_f^{1/2} Q^{1/2} n^{-1/2} + p_f Q n^{-1} + Q^{-1}). \quad (\text{S.15})$$

We find the rate of I_3^{PP} now. Using $g(0) = g(1) = 0$, we have

$$\begin{aligned} \bar{\mathbf{E}}(s) &= \sum_{l=1}^{Q-1} g \left(\frac{l}{Q} \right) (\mathbf{E}(s+l) - \mathbf{E}(s+l-1)) \\ &= \sum_{l=0}^{Q-1} \left[g \left(\frac{l}{Q} \right) - g \left(\frac{l+1}{Q} \right) \right] \mathbf{E}(s+l) = -Q^{-1} \sum_{l=0}^{Q-1} g'_{e_{l+1}} \mathbf{E}(s+l), \end{aligned}$$

where $l/Q \leq e_{l+1} \leq (l+1)/Q$. Hence

$$\begin{aligned} \mathbf{p}_{ij}^T \bar{\mathbf{E}}(s) \bar{\mathbf{E}}(s)^T \mathbf{p}_{ij} &= \mathbf{p}_{ij}^T \left(Q^{-1} \sum_{l=0}^{Q-1} g'_{e_{l+1}} \mathbf{E}(s+l) \right) \left(Q^{-1} \sum_{l=0}^{Q-1} g'_{e_{l+1}} \mathbf{E}(s+l)^T \right) \mathbf{p}_{ij} \\ &= Q^{-2} \left(\sum_{l=0}^{Q-1} g_{e_{l+1}}'^2 \mathbf{p}_{ij}^T \mathbf{E}(s+l) \mathbf{E}(s+l)^T \mathbf{p}_{ij} + \sum_{l \neq l'} g'_{e_{l+1}} g'_{e_{l'+1}} \mathbf{p}_{ij}^T \mathbf{E}(s+l) \mathbf{E}(s+l')^T \mathbf{p}_{ij} \right). \end{aligned} \quad (\text{S.16})$$

By Lemma 4 and Lemma 5, we then have

$$\begin{aligned} & \left| \frac{I_3^{PP}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right| \\ &= O_P \left(Q^{-1} Q^{-2} \left(\sum_{l=0}^{Q-1} \sum_{s=1}^{n-Q+1} \frac{\mathbf{p}_{ij}^T (\mathbf{E}(s+l) \mathbf{E}(s+l)^T) \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} + 2 \sum_{l' < l} \sum_{s=1}^{n-Q+1} \frac{\mathbf{p}_{ij}^T (\mathbf{E}(s+l) \mathbf{E}(s+l')^T) \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right) \right) \\ &= O_P(Q^{-2}(n) + Q^{-3/2}(n^{1/2})) = O_P(Q^{-2}n). \end{aligned} \quad (\text{S.17})$$

For $I_2^{PP}/\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}$, it can be decomposed as $\sum_{i=1}^6 L_i/\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}$, where

$$\begin{aligned} L_1 &= \frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} \sum_{l=0}^{Q-1} g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^T (\mathbf{X}_{v_{s+l}} - \mathbf{X}_{v_{s+l-1}}) \sum_{l'=0}^{Q-1} (-g_{e_{l'+1}}') Q^{-1} \mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s+l'), \\ L_2 &= \frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} \sum_{l=0}^{Q-1} g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^T (\mathbf{X}_{v_{s+l}} - \mathbf{X}_{v_{s+l-1}}) \sum_{l'=0}^{Q-1} (-g_{e_{l'+1}}') Q^{-1} \mathbf{p}_{ij}^T (\mathbf{X}(s+l') - \mathbf{X}_{v_{s+l'}}), \\ L_3 &= \frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} \sum_{l=0}^{Q-1} g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^T (\mathbf{X}_{v_{s+l}} - \mathbf{X}_{v_{s+l-1}}) \sum_{l'=0}^{Q-1} (-g_{e_{l'+1}}') Q^{-1} \mathbf{p}_{ij}^T (e(\mathbf{J}(s+l')) - e(\mathbf{J}_{v_{s+l'}})), \\ L_4 &= \frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} \sum_{l=0}^{Q-1} g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^T (e(\mathbf{J}_{v_{s+l}}) - e(\mathbf{J}_{v_{s+l-1}})) \sum_{l'=0}^{Q-1} (-g_{e_{l'+1}}') Q^{-1} \mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s+l'), \\ L_5 &= \frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} \sum_{l=0}^{Q-1} g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^T (e(\mathbf{J}_{v_{s+l}}) - e(\mathbf{J}_{v_{s+l-1}})) \sum_{l'=0}^{Q-1} (-g_{e_{l'+1}}') Q^{-1} \mathbf{p}_{ij}^T (\mathbf{X}(s+l') - \mathbf{X}_{v_{s+l'}}), \\ L_6 &= \frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} \sum_{l=0}^{Q-1} g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^T (e(\mathbf{J}_{v_{s+l}}) - e(\mathbf{J}_{v_{s+l-1}})) \sum_{l'=0}^{Q-1} (-g_{e_{l'+1}}') Q^{-1} \mathbf{p}_{ij}^T (e(\mathbf{J}(s+l')) - e(\mathbf{J}_{v_{s+l'}})). \end{aligned}$$

Using notations as before, we have

$$\begin{aligned} \sum_{l=0}^{Q-1} g\left(\frac{l}{Q}\right) \frac{\mathbf{p}_{ij}^T (\mathbf{X}_{v_{s+l}} - \mathbf{X}_{v_{s+l-1}})}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} &= \sum_{l=0}^{Q-1} \frac{A_s(l) + V_s(l)}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} = O_P(p_f^{1/2} Q n^{-1} + Q^{1/2} n^{-1/2}), \\ \sum_{l=0}^{Q-1} \frac{(-g_{e_{l+1}}') Q^{-1} \mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s+l)}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} &= O_P(Q \cdot Q^{-2} + Q^2 \cdot Q^{-2} n^{-1})^{1/2} = O_P(Q^{-1/2}), \\ \sum_{l=0}^{Q-1} (-g_{e_{l+1}}') Q^{-1} \frac{\mathbf{p}_{ij}^T (\mathbf{X}(s+l) - \mathbf{X}_{v_{s+l'}})}{(\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^{1/2}} &= O_P(p_f n^{-1} + Q^{-1} Q^{1/2} n^{-1/2}) = O_P(p_f^{1/2} n^{-1} + Q^{-1/2} n^{-1/2}). \end{aligned}$$

With these, together with Assumption (W1) to (W3) and the rate of $n^{-1/4}$ in Fan and Wang (2007) for jump removal, we have

$$\begin{aligned} L_3/\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij} &= O_P((p_f^{1/2} Q n + Q^{1/2} n^{-1/2}) \cdot Q^{-1} n^{-1/4}) = O_P(p_f^{1/2} n^{-5/4} + Q^{-1/2} n^{-3/4}), \\ L_4/\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij} &= O_P(Q^{-1} \cdot Q^{-1/2} \cdot Q n^{-1/4}) = O_P(Q^{-1/2} n^{-1/4}), \\ L_5/\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij} &= O_P(Q^{-1} Q n^{-1/4} (p_f^{1/2} n^{-1} + Q^{-1/2} n^{-1/2})) = O_P(p_f^{1/2} n^{-5/4} + Q^{-1/2} n^{-3/4}), \\ L_6/\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij} &= O_P(Q^{-1} \cdot Q n^{-1/4} \cdot n^{-1/4}) = O_P(n^{-1/2}). \end{aligned}$$

For L_1 , consider $L_1 = \sum_{l=0}^{Q-1} L_{1,l'}$, where

$$L_{1,l'} = \frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} \sum_{l=0}^{Q-1} g\left(\frac{l}{Q}\right) \mathbf{p}_{ij}^T (\mathbf{X}_{v_{s+l}} - \mathbf{X}_{v_{s+l-1}}) (-g_{e'_{l'+1}}) Q^{-1} \mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s+l').$$

Then using Assumption (E3),

$$E_j(L_{1,l'}^2 / (\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij})^2) = O(Q^{-2} \cdot n \cdot (p_f^{1/2} Q n^{-1} + Q^{1/2} n^{-1/2})^2 Q^{-2}) = O(p_f Q^{-2} n^{-1} + Q^{-3}).$$

Hence

$$L_1/\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij} = O_P(p_f^{1/2} n^{-1/2} + Q^{-1/2}).$$

We also have

$$\begin{aligned} L_2/\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij} &= O_P(Q^{-1} n (p_f Q n^{-1} + Q^{1/2} n^{-1/2}) (p_f^{1/2} n^{-1} + Q^{-1/2} n^{-1/2})) \\ &= O_P(p_f n^{-1} + Q^{-1}). \end{aligned}$$

Hence we have

$$\left| \frac{I_2^{PP}}{\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right| = O_P(p_f^{1/2} n^{-1/2} + Q^{-1/2}). \quad (\text{S.18})$$

The above results imply that

$$\left| \frac{\mathbf{p}_{ij}^T \mathbf{P}(\tilde{\mathbf{Y}})_j \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1 \right| = O_P(Q^{-1/2} + Q^{-2} n + p_f^{1/2} Q^{1/2} n^{-1/2} + p_f Q n^{-1}).$$

When there are no factors, $p_f = 1$, and so the best rate is to balance $Q^{-2} n$ and $Q^{1/2} n^{-1/2}$, resulting in $Q \asymp n^{3/5}$ and a rate of convergence at $n^{-1/5}$. When there are pervasive factors such that $p_f = p$, then we need $Q \asymp n^{3/5} p^{-1/5}$, and rate of convergence at $n^{-1/5} p^{2/5}$. Hence we need $p = o(n^{1/2})$ for guaranteed convergence in this case.

This completes the proof for the positive semi-definite version, since the above rate does not depend on both the indices i and j .

Consider now NER-PRVM which is $\tilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)^P$. Using the same decomposition as $\mathbf{P}(\tilde{\mathbf{Y}})_j$, we have

$$\begin{aligned} \mathbf{p}_{ij}^T \tilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)^P \mathbf{p}_{ij} &= \mathbf{p}_{ij}^T \mathbf{P}(\tilde{\mathbf{Y}})_j \mathbf{p}_{ij} - \varsigma(\psi Q)^{-1} \sum_{s=1}^{n-Q+1} \mathbf{p}_{ij}^T \hat{\boldsymbol{\eta}}^{(j)} \mathbf{p}_{ij}, \\ &= I_1^P + 2I_2^P + I_3^P, \end{aligned}$$

where I_1^P , and I_2^P are exactly the same as I_1^{PP} and I_2^{PP} from the decomposition of $\mathbf{p}_{ij}^T \mathbf{P}(\tilde{\mathbf{Y}}) \mathbf{p}_{ij}$ at the beginning of the proof, and

$$I_3^P = \frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} [\mathbf{p}_{ij}^T \bar{\mathbf{E}}(s) \bar{\mathbf{E}}(s)^T \mathbf{p}_{ij} - \varsigma \cdot \mathbf{p}_{ij}^T \hat{\boldsymbol{\eta}}^{(j)} \mathbf{p}_{ij}].$$

From (S.15) and (S.18), we have

$$\left| \frac{I_1^P}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1 \right| = O_P(p_f^{1/2} Q^{1/2} n^{-1/2} + p_f Q n^{-1} + Q^{-1}),$$

$$\left| \frac{I_2^P}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right| = O_P(p_f^{1/2} n^{-1/2} + Q^{-1/2}).$$

We can decompose $I_3^P = J_1 + J_2$, where

$$J_1 = \frac{1}{\psi Q} \sum_{s=1}^{n-Q+1} [\mathbf{p}_{ij}^T \bar{\mathbf{E}}(s) \bar{\mathbf{E}}(s)^T \mathbf{p}_{ij} - \varsigma \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij}],$$

$$J_2 = \frac{\varsigma}{\psi Q} \sum_{s=1}^{n-Q+1} (\mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij} - \mathbf{p}_{ij}^T \hat{\boldsymbol{\eta}}^{(j)} \mathbf{p}_{ij}).$$

Since $\bar{\mathbf{E}}(s) = \sum_{l=1}^{Q-1} [g(l/Q) - g((l+1)/Q)] \mathbf{E}(s+l) = Q^{-1} \sum_{l=1}^{Q-1} (-g'_{e_{l+1}}) \mathbf{E}(s+l)$, we have

$$J_1 = J_{1,1} + J_{1,2}, \text{ where}$$

$$J_{1,1} = \frac{1}{\psi Q} \sum_{l=1}^{Q-1} \left[g\left(\frac{l}{Q}\right) - g\left(\frac{l+1}{Q}\right) \right]^2 \sum_{s=1}^{n-Q+1} (\mathbf{p}_{ij}^T \mathbf{E}(s+l) \mathbf{E}(s+l)^T \mathbf{p}_{ij} - \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij}),$$

$$J_{1,2} = \psi^{-1} Q^{-3} \sum_{l \neq l'} g'_{e_{l+1}} g'_{e_{l'+1}} \sum_{s=1}^{n-Q+1} \mathbf{p}_{ij}^T \mathbf{E}(s+l) \mathbf{E}(s+l')^T \mathbf{p}_{ij}.$$

From (S.17), we know that

$$J_{1,2} / \mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij} = O_P(Q^{-3/2} n^{1/2}). \quad (\text{S.19})$$

To find the rate of $J_{1,1}$, for $l < Q$, it suffice to consider the rate of

$$\sum_{s=1}^{n-Q+1} \frac{\mathbf{p}_{ij}^T \mathbf{E}(s+l) \mathbf{E}(s+l)^T \mathbf{p}_{ij} - \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} = E'_{1,1} + E'_{1,2} + E_2 + E_3 + 2 \sum_{k=4}^6 E_k,$$

where E_2 to E_6 are essentially the same as those defined in the proof of Lemma 4 (only with s replaced by $s+l$ and n by $n-Q+1$), and

$$E'_{1,1} = \sum_{s=1}^{n-Q+1} \frac{\mathbf{p}_{ij}^T \boldsymbol{\epsilon}(s+l) \boldsymbol{\epsilon}(s+l)^T \mathbf{p}_{ij} - \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s+l}^j \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}},$$

$$E'_{1,2} = \sum_{s=1}^{n-Q+1} \frac{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s+l}^j \mathbf{p}_{ij} - \mathbf{p}_{ij}^T \boldsymbol{\Sigma}_{\epsilon, s}^j \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}.$$

Under Assumption (E3), using Lemma 2.7 of Bai and Silverstein (1998), we have $E_j(E'_{1,1}) = 0$, and

$$E_j((E'_{1,1})^2 | \Sigma_{\epsilon,u}, u \in [0, 1]) = O(n + n^2 \cdot n^{-1}) = O(n).$$

Hence

$$E'_{1,1} = O_P(n^{1/2}).$$

Also, by Assumption (E1) on the smoothness of $\{\Sigma_{\epsilon,s}\}_s$, we have

$$E'_{1,2} = O_P(|l|).$$

From the proof of Lemma 4, we have

$$E_2 = O_P(1), \quad E_3 = O_P(n^{-1/2}).$$

Using Assumption (W1) to (W3) and the jump removal rate of $n^{-1/4}$ from Fan and Wang (2007),

$$E_5 = O_P(n^{-1/4}), \quad E_6 = O_P(n^{-3/4}).$$

For E_4 , using Assumption (E3),

$$E_4 = O_P(n \cdot (n^{-1/2})^2 + n^2 \cdot n^{-1} \cdot (n^{-1/2})^2) = O_P(1).$$

From these rates, we have

$$\frac{J_{1,1}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} = O_P\left(\frac{1}{\psi Q} \sum_{l=1}^{Q-1} (g'_{el+1})^2 Q^{-2} (n^{1/2} + l)\right) = O_P(Q^{-1} + Q^{-2} n^{1/2}).$$

Hence

$$J_1 / \mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij} = O_P(Q^{-1} + Q^{-3/2} n^{1/2}) = O_P(Q^{-3/2} n^{1/2}).$$

For J_2 , consider

$$\begin{aligned} \frac{\mathbf{p}_{ij}^T \hat{\boldsymbol{\eta}}^{(j)} \mathbf{p}_{ij} - \mathbf{p}_{ij}^T \Sigma_{\epsilon,s}^j \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} &= \sum_{i=1}^5 R_i, \text{ where} \\ R_1 &= \frac{1}{2n} \sum_{s=2}^n \frac{\mathbf{p}_{ij}^T \mathbf{E}(s) \mathbf{E}(s)^T \mathbf{p}_{ij} - \mathbf{p}_{ij}^T \Sigma_{\epsilon,s}^j \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - \frac{\mathbf{p}_{ij}^T \Sigma_{\epsilon,s}^j \mathbf{p}_{ij}}{2n \mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ R_2 &= \frac{1}{2n} \sum_{s=2}^n \frac{\mathbf{p}_{ij}^T \mathbf{E}(s-1) \mathbf{E}(s-1)^T \mathbf{p}_{ij} - \mathbf{p}_{ij}^T \Sigma_{\epsilon,s}^j \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - \frac{\mathbf{p}_{ij}^T \Sigma_{\epsilon,s}^j \mathbf{p}_{ij}}{2n \mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ R_3 &= \frac{-1}{n} \sum_{s=2}^n \frac{\mathbf{p}_{ij}^T \mathbf{E}(s) \mathbf{E}(s-1)^T \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ R_4 &= \frac{1}{2n} \sum_{s=2}^n \frac{\mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_s} - \tilde{\mathbf{X}}_{v_{s-1}})(\tilde{\mathbf{X}}_{v_s} - \tilde{\mathbf{X}}_{v_{s-1}})^T \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}, \\ R_5 &= \frac{1}{n} \sum_{s=2}^n \frac{\mathbf{p}_{ij}^T (\tilde{\mathbf{X}}_{v_s} - \tilde{\mathbf{X}}_{v_{s-1}})(\mathbf{E}(s) - \mathbf{E}(s-1))^T \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}}. \end{aligned}$$

From the proof of the rate for J_1 before, and Assumption (E1), we get

$$R_1, R_2 = O_P(n^{-1} \cdot n^{1/2} + n^{-1}) = O_P(n^{-1/2}).$$

From Lemma 5, we have

$$R_3 = O_P(n^{-1} \cdot n^{1/2}) = O_P(n^{-1/2}).$$

From the result of Lemma 1, we immediately have

$$R_4 = O_P(n^{-1}(n^{-1/2} + p_f^{1/2}n^{-1/2}) + n^{-1}) = O_P(n^{-1}).$$

Substituting $K_m = 1$ in (S.5) and incorporating Assumption (A4), we have

$$R_5 = O_P(n^{-1} \cdot 1) = O_P(n^{-1}).$$

These rates imply that, with $\varsigma = O(Q^{-1})$,

$$J_2 = O_P(Q^{-2} \cdot n \cdot n^{-1/2}) = O_P(Q^{-2}n^{1/2}).$$

Finally we have

$$\left| \frac{I_3^P}{\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right| = O_P(Q^{-3/2}n^{1/2} + Q^{-2}n^{1/2}) = O_P(Q^{-3/2}n^{1/2}).$$

Hence

$$\begin{aligned} & \max_{\substack{i=1,\dots,p \\ j=1,\dots,L}} \left| \frac{\mathbf{p}_{ij}^T \tilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)^P \mathbf{p}_{ij}}{\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1 \right| \\ & \leq \max_{\substack{i=1,\dots,p \\ j=1,\dots,L}} \left| \frac{I_1^P}{\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} - 1 \right| + 2 \max_{\substack{i=1,\dots,p \\ j=1,\dots,L}} \left| \frac{I_2^P}{\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right| + \max_{\substack{i=1,\dots,p \\ j=1,\dots,L}} \left| \frac{I_3^P}{\mathbf{p}_{ij}^T \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{p}_{ij}} \right| \\ & = O_P(p_f^{1/2}Q^{1/2}n^{-1/2} + Q^{-1/2} + Q^{-3/2}n^{1/2}). \end{aligned}$$

When there are no factors such that $p_f = 1$, the above rates imply that $Q \asymp n^{1/2}$, with a rate of convergence at $n^{-1/4}$. When there are pervasive factors such that $p_f = p$, the best rate can be achieved with $Q \asymp n^{1/2}p^{-1/4}$, with rate of convergence at $p^{3/8}n^{-1/4}$. So we need $p = o(n^{2/3})$ for guaranteed convergence. This completes the proof for the bias-corrected version.

Finally, using similar arguments at the end of the proof of Theorem 1 in Section , we have

$$\|\hat{\mathbf{\Sigma}}(0, 1)^P \mathbf{\Sigma}_{\text{Ideal}}(0, 1)^{-1} - \mathbf{I}_p\| = O_P(p_f^{1/2}Q^{1/2}n^{-1/2} + Q^{-1/2} + Q^{-3/2}n^{1/2}),$$

the same as for $\hat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)^P$. Similar treatments apply for $\hat{\mathbf{\Sigma}}(0, 1)^{PP}$. This completes the proof of the theorem. $\square$

Proof of the Theorem 4

We have shown in the proofs of Theorem 1 to 3 for the results using jumps-removed data. To complete the proof, note that

$$\begin{aligned} \left\| \sum_{0 \leq t \leq 1} (\Delta \mathbf{J}_t \Delta \mathbf{J}_t^T - \Delta \hat{\mathbf{J}}_t \Delta \hat{\mathbf{J}}_t^T) \right\| & \leq C \max_{0 \leq t \leq 1} \|\Delta \mathbf{J}_t \Delta \mathbf{J}_t^T - \Delta \hat{\mathbf{J}}_t \Delta \hat{\mathbf{J}}_t^T\| \\ & \leq 2C \max_{0 \leq t \leq 1} \|\Delta \mathbf{J}_t - \Delta \hat{\mathbf{J}}_t\| \cdot \|\Delta \mathbf{J}_t\| + C \max_{0 \leq t \leq 1} \|\Delta \mathbf{J}_t - \Delta \hat{\mathbf{J}}_t\|^2 \\ & = O_P(n^{-1/4}L^{-1/4}). \end{aligned}$$

Since L is finite, this completes the proof of the theorem. $\square$

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